Geometric Spatial Prior-Guided Dynamic Graph Learning With Fourier-Modulated Superpixels for Hyperspectral Image Classification

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***Abstract*—Superpixel-based graph methods are efficient for hyperspectral image (HSI) classification (HSIC), yet they often rely on static graph topologies fixed during preprocessing. These predetermined structures, based primarily on spatial adjacency, generally fail to capture the richer semantic correlations present in HSI data. Moreover, superpixel node features are often obtained by simple averaging, making them susceptible to mixed noise and spectral heterogeneity. To overcome these limitations, we propose a novel framework termed GSDG-Net, which integrates geometry-guided dynamic graphs with Fourier-modulated superpixels. Instead of using a fixed adjacency graph, our approach first prunes long-range spatial connections based on centroid distances to obtain a relatively sparse prior graph. It subsequently applies a content-aware attention mechanism to dynamically reweight the remaining edges, explicitly prioritizing semantic affinity to construct a semantically aware dynamic graph. To address the issue of feature ambiguity, we devise an innovative Fourier-domain spectral modulation (FDSM) module. A differentiable complex-valued Fourier-domain gating mechanism lies at the heart of this module. It acts as an adaptive, learnable spectral filter that emphasizes informative low-frequency components and attenuates high-frequency noise, directly yielding discriminative Fourier-modulated superpixel features. Further augmenting the representation, a multiscale convolutional branch compensates for the pixel-level detail loss that arises from superpixel abstraction. Experiments on four benchmark HSI datasets demonstrate that GSDG-Net achieves state-of-the-art performance under few-shot settings. The code is available at https://github.com/LanLanZhao/IEEE_TGRS_GSDG-Net.**

***Index Terms*—Convolutional neural network (CNN), Fourier transform, graph attention network (GAT), hyperspectral image classification (HSIC), superpixel segmentation.**

I. Introduction

HYPERSPECTRAL imagery records a dense set of contiguous spectral measurements for each spatial location, preserving subtle absorption patterns that are highly informative for distinguishing categories with very similar appearances. Due to this discriminative capability, hyperspectral image classification (HSIC) [1] has become a key component in diverse remote sensing applications, ranging from land-cover mapping [2], [3] and agricultural analysis [4] to mineral exploration [5], [6], [7] and ecological monitoring [8], [9], [10]. Despite their potential, hyperspectral scenes are characterized by strong spectral redundancy [11], mixed pixels [12], and complex spatial layouts [13]. These intrinsic complexities are further exacerbated by the high cost of pixel-wise annotation, creating a significant gap between available data volume and labeling budgets [14]. Consequently, developing accurate HSIC models under limited supervision [15] has emerged as a critical research challenge that requires both computational efficiency and robustness to spectral noise.

In recent years, deep learning (DL) [16] has been widely introduced into HSIC and has shown clear advantages over conventional pipelines relying on handcrafted descriptors. Benefiting from hierarchical representation learning, DL models can directly optimize discriminative spectral–spatial features [17] from raw data in an end-to-end manner.

Convolutional neural networks (CNNs) [18], [19] have become a mainstream choice for HSIC, as local receptive fields and weight sharing provide an effective inductive bias for learning discriminative spectral–spatial patterns with moderate computational cost. Early pipelines mainly exploited spectral cues via 1-D convolutions [20], [21], improving pixel-wise recognition but often underutilizing spatial textures. Later studies extended CNNs toward 2-D/3-D [22], [23], [24] formulations to jointly encode neighborhood structure and spectral variation. To enable deeper architectures without degradation, Zhong et al. [25] introduced residual learning to propose the spectral–spatial residual network (SSRN), which stabilizes optimization for hierarchical abstraction. Building on this, Fu et al. [26] developed the local–global gating CNN (LGG-CNN) by inserting gating blocks to refine representations via signals from varying receptive fields. In parallel, aiming for lightweight efficiency, Ma et al. [27] designed the ultralight hybrid CNN (ULite-R2HCN) by removing redundancy in convolutions; it employs structured multiscale 3-D kernels and a copy–splicing–grouping strategy to capture nonneighboring correlations. To improve representation discriminability under limited labeled data, Liu et al. [28] developed a CNN-based deep contrastive learning network for small-sample HSIC by combining convolutional feature extraction with contrastive learning and pseudo-label refinement. Despite these advances, CNN-based methods still limit

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long-range dependency modeling due to fixed kernel geometry; simply stacking layers to expand the receptive field often increases computational overhead and overfitting risk.

To break the bottleneck of restricted receptive fields in CNNs, recent research has pivoted toward Transformer-based architectures. Unlike static convolutions, these models employ dynamic self-attention mechanisms to explicitly capture long-range dependencies. Pioneering this direction, Hong et al. [29] proposed SpectralFormer, which rethinks HSIC from a sequence modeling perspective to learn discriminative spectral–spatial representations. Aiming to mitigate the high computational cost of standard self-attention, Wang et al. [30] developed the spatial–spectral fusion BiFormer (SSFBF). This method adopts bi-level dynamic routing to reduce redundant token interactions and incorporates spectral fusion to enhance cross-layer aggregation. However, acknowledging that pure Transformers may lack the local inductive bias inherent in convolutions, recent works have shifted toward hybrid CNN–Transformer architectures to synergize global context with local features. In this context, Yang et al. [31] proposed the adaptive coupling Transformer network (ACTN), which constructs dual branches to extract local contextual and global Transformer features while using adaptive reciprocal fusion to promote information exchange. To further facilitate efficient interactions across multiple dimensions, Zhou et al. [32] presented the dual-branch convolutional Transformer network (DCTN), where an efficient self-attention module enables interactions across height, width, and spectral domains, complemented by a convolutional branch. Pushing this hybrid paradigm to a finer granularity, Chen et al. [33] developed the multigranularity spatial–spectral convolution Transformer (MSSCFormer). It integrates intraclass spectral attention to reduce variability and employs a global–local block to seamlessly combine attention-based global modeling with convolutional refinement.

In recent years, graph neural networks (GNNs) have emerged as a powerful paradigm for HSIC, transcending the limits of regular grids to model the intrinsic non-Euclidean structures of land covers [34]. By constructing topological graphs where nodes represent pixels or superpixels, GNNs perform effective relational reasoning over irregular regions. Among the diverse architectures tailored for this task, graph convolutional networks (GCNs) [35] and graph attention networks (GATs) [36] have become mainstream choices for feature aggregation, while graph autoencoders (GAEs) [37] are widely explored for unsupervised representation learning. Early studies predominantly adopted pixel-wise graph construction strategies. As a pioneering attempt, Shahraki and Prasad [38] introduced GCNs into HSIC, facilitating low-dimensional feature representation. To enhance feature expressiveness, Qin et al. [39] improved graph construction by explicitly integrating spectral and spatial cues, while Mou et al. [40] developed nonlocal graph convolution to strengthen contextual modeling beyond local neighborhoods. However, the prohibitively high computational cost of pixel-wise graphs spurred the transition toward superpixel-based alternatives. Focusing on hierarchical representation within this efficient paradigm, Liu et al. [41] proposed the multilevel superpixel structured graph U-Nets (MSSGU) to capture multiscale context. Pushing this paradigm further, Zou et al. [42] introduced PSFormer, which employs a progressive superpixel merging strategy to efficiently aggregate local–global features. From an optimization perspective, Xue et al. [43] developed DSR-GCN to enhance graph reasoning via multiscale constraints. However, these architectures operate primarily as bottom-up encoders. They effectively abstract information but lack an explicit top-down decoding mechanism to restore fine-grained pixel details from coarse nodes. More recently, recognizing that superpixel-based GNNs may sacrifice fine-grained local details, hybrid architectures fusing CNNs and GNNs have gained traction. Addressing the fundamental structural misalignment between regular image grids and irregular graph topologies, Liu et al. [44] pioneered the CNN-enhanced GCN (CEGCN). Crucially, they designed a graph encoder–decoder mechanism to enable seamless feature propagation between pixel-level CNNs and superpixel-level GNNs, establishing a robust foundation for cross-domain fusion. Following this hybrid fusion paradigm, Yu et al. [45] proposed a two-branch deeper GCN to strengthen structural representation learning through an improved graph architecture, while using a CNN branch to preserve fine-grained spectral–spatial features. Subsequently, Zhou et al. [46] proposed AMGCFN, which employs parallel CNN and GCN branches to jointly extract hierarchical features. To further enrich structural modeling, Liu et al. [47] developed STSM-DGCN, combining a spatial graph branch with a spectral multigraph branch. Pushing topology modeling beyond pairwise connections, Wang et al. [48] introduced HGCN-MHF, integrating CNNs with hypergraph convolution to exploit high-order correlations.

Despite recent advances in superpixel-based graph methods, several critical challenges remain. First, graph topologies are typically constructed statically during preprocessing, based primarily on spatial adjacency while ignoring semantic similarity [44], [49]. This fixed structure limits information flow to local neighbors, preventing effective fusion of semantically consistent features across nonadjacent regions. Second, node features are usually obtained by simple averaging or pooling of pixel-level responses within each superpixel. Such aggregation fails to account for spectral variability and mixed noise inside superpixels, leading to ambiguous representations. These noisy features can distort downstream graph learning and message passing, often propagating irrelevant noise instead of meaningful class signals.

To effectively overcome these limitations, we propose a novel framework termed GSDG-Net, which synergizes geometry-guided dynamic graph learning with Fourier-modulated superpixels. Unlike conventional methods constrained by static superpixel adjacency matrices predefined solely by spatial proximity, our approach devises a geometry-aware dynamic graph mechanism. It transcends fixed connectivity patterns by adaptively refining graph adjacency based on both spatial layout and intrinsic semantic content. Specifically, we first construct a sparse prior based on centroid proximity and subsequently refine it via a content-aware attention module to capture intrinsic semantic correlations within an expanded spatial neighborhood. Crucially, to fundamentally mitigate node feature ambiguity, we introduce an innovative Fourier-domain spectral modulation (FDSM) module. This module transforms node features into the frequency domain and employs a learnable complex-valued gating mechanism

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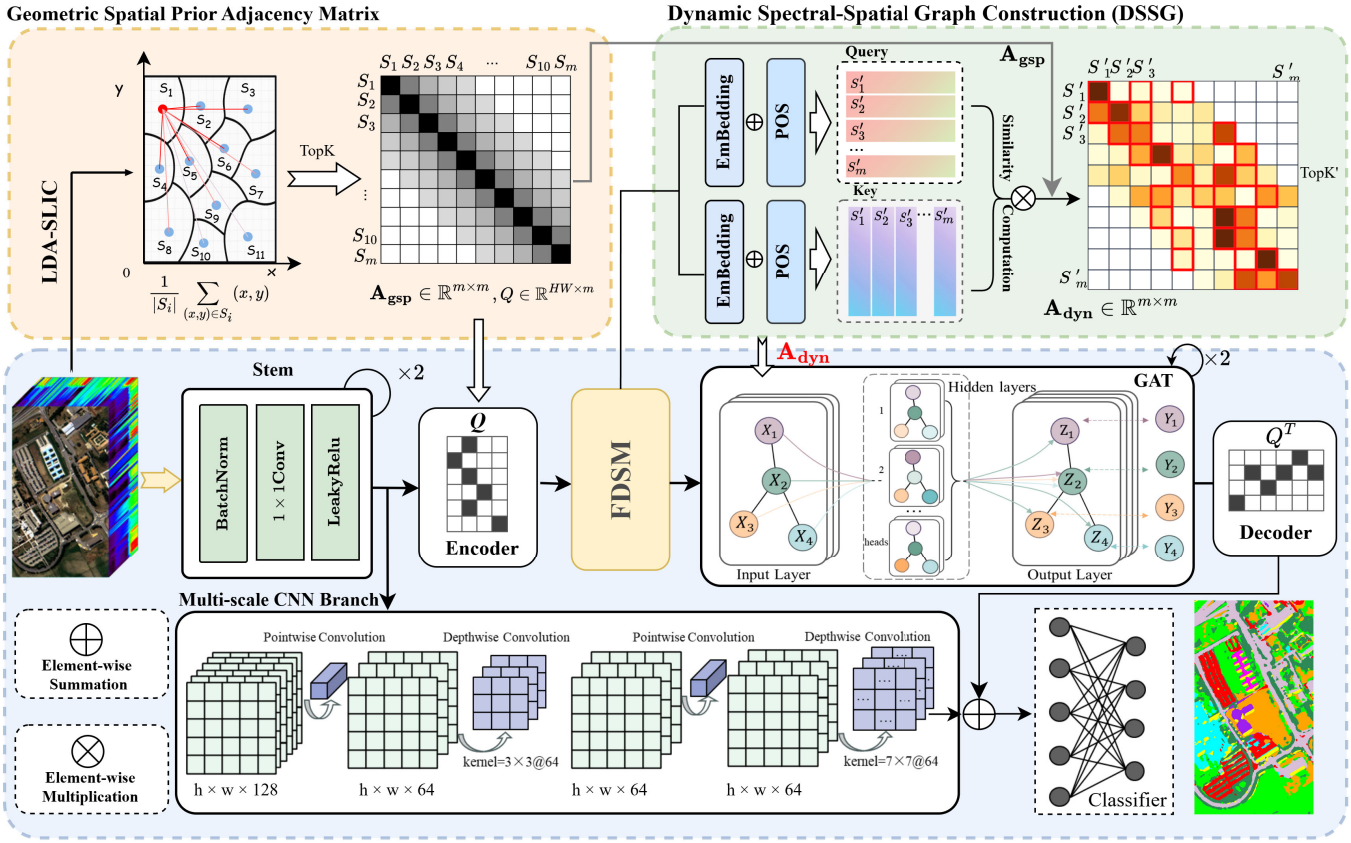


Fig. 1. Framework of the GSDG-Net model.

to jointly modulate both amplitude and phase information. This design effectively isolates informative spectral components while suppressing background noise, thereby ensuring high-fidelity feature representations for subsequent reasoning. Finally, to compensate for the graph branch's limited capacity in modeling fine-grained pixel-level details, we incorporate a cascaded progressive multiscale convolutional structure. This architecture progressively expands the receptive field across layers to effectively capture multiscale spatial context at the pixel level, thereby complementing the graph branch with fine-grained local detail for holistic scene understanding.

1) We propose GSDG-Net, a novel learnable superpixel-graph framework for HSIC. It synergizes Fourier-domain node purification and geometry-aware dynamic graph learning in an end-to-end manner, replacing the static superpixel adjacency graph with adaptive connectivity.

2) We innovatively propose a coarse-to-fine dynamic graph construction strategy. It leverages superpixel centroids to define a broad candidate search space, which is then adaptively pruned by content-aware attention to ensure robust connectivity among semantically homogeneous regions.

3) We design a novel FDSM module for feature refinement. It transforms node features into the Fourier domain and utilizes a learnable complex-valued gate to jointly modulate amplitude and phase, thereby suppressing aliasing noise while emphasizing discriminative features.

4) Extensive experiments on four widely used HSI benchmarks [Indian Pines (IP), University of Pavia (PU), Salinas, and Utopia Planitia (UP)] demonstrate that GSDG-Net consistently achieves state-of-the-art performance in few-shot HSIC, validating its effectiveness and robustness.

II. Method

As shown in Fig. 1, GSDG-Net integrates a geometrically constrained dynamic graph branch and a multiscale convolutional module to jointly model the regional semantics at the superpixel level and the multiscale spatial details at the pixel level in hyperspectral images (HSIs). The proposed framework initially employs Fourier-domain spectral modulation (FDSM) to refine aggregated superpixel nodes. While superpixel aggregation yields compact regional representations, it often introduces mixed spectral noise and boundary artifacts. By purifying features at this stage, FDSM not only strengthens node representations but also furnishes robust semantic cues for subsequent dynamic graph construction. Since adaptive edge generation relies on these refined features, the enhanced representations directly govern edge reweighting and neighbor selection, thereby fostering a more discriminative and stable graph topology. Finally, graph structure learning is formulated as a geometry-regularized semantic search, where connections are dynamically established based on spectral similarity within spatially plausible neighborhoods constrained by geometric priors. Finally, the region-level representations from the graph branch are fused with the pixel-level features extracted by the convolutional branch, effectively bridging the scale gap between different granularities. We next describe the key components of the proposed framework in detail.

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A. Superpixel Generation and the Geometric Spatial Prior Adjacency Matrix

Direct pixel-wise graph construction on HSIs is computationally prohibitive and sensitive to spectral noise due to their high spectral dimensionality and spatial redundancy. To address this, we adopt a superpixel-based approach to generate compact graph nodes and initialize a spatial geometric prior, which serves as a positional guide for the subsequent dynamic graph learning.

1) Discriminative Superpixel Generation: To ensure that superpixels capture semantically meaningful regions rather than relying solely on raw color similarity, we first employ linear discriminant analysis (LDA) [50] to project the original HSI cube $\mathbf{X}_{raw} \in \mathbb{R}^{H\times W\times B}$ into a compact discriminative subspace. This yields $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ with $C \ll B$, which enhances class separability while preserving spatial structures.

Then, we apply the simple linear iterative clustering (SLIC) [51] algorithm on $\mathbf{X}$ to partition the image into N nonoverlapping superpixels $\mathcal{S} = \{S_1, S_2, \ldots, S_N\}$. Specifically, we construct a pixel-to-superpixel association matrix $\mathbf{Q} \in \mathbb{R}^{HW\times N}$ that links each pixel to its corresponding superpixel

$$Q_{i,j} = \begin{cases} 1, & \text{if } \hat{\mathbf{x}}_i \in S_j \\ 0, & \text{otherwise} \end{cases} \qquad \hat{\mathbf{X}} = \text{Flatten}(\mathbf{X}) \tag{1}$$

where Flatten$(\cdot)$ denotes flattening $\mathbf{X}$ along the spatial dimensions, and $\hat{\mathbf{x}}_i \in \mathbb{R}^C$ is the ith pixel feature in $\hat{\mathbf{X}}$.

To obtain superpixel-level node representations, we further perform a column-wise normalization on $\mathbf{Q}$ as $\hat{\mathbf{Q}} = \mathbf{Q}\mathbf{D}^{-1}$, where $\mathbf{D} = \text{diag}(d_1, \ldots, d_N)$ and $d_j = \sum_{i=1}^{HW} Q_{i,j}$. Then, an encoder-style aggregation is defined to convert pixel-wise features into superpixel node features

$$\mathbf{H}_{init} = \text{Encoder}(\mathbf{X}; \mathbf{Q}) = \hat{\mathbf{Q}}^{\top}\hat{\mathbf{X}} \tag{2}$$

where $\mathbf{H}_{init} \in \mathbb{R}^{N\times C}$ denotes the initial node feature matrix. In essence, (2) computes the mean feature of each superpixel by averaging all pixels within S_j, yielding robust superpixel descriptors that will be further refined by the subsequent Fourier-domain modulation module.

2) Construction of the Geometric Spatial Prior Adjacency Matrix: Most existing superpixel-based GNNs typically adopt a static local adjacency strategy, allowing connections only between superpixels that share a common boundary. This rigid topological constraint severely restricts the receptive field, confining information propagation strictly within local neighborhoods and thereby failing to effectively capture long-range dependencies that are spatially discontinuous, yet semantically related. To overcome this limitation, we introduce a geometric spatial prior adjacency matrix. Its core idea is to construct a flexible candidate connectivity domain for each superpixel node, rather than directly determining connections through rigid rules. Specifically, we generate this matrix based on the relative distances between the spatial centroids of superpixels, thereby establishing an extensive geometric neighborhood that transcends the constraint of enforcing edges only between immediately adjacent superpixels.

Given the irregular shapes of superpixels, we employ the spatial centroid as a geometric proxy to standardize the distance calculation. Let Ω_i denote the set of pixel indices belonging to the ith superpixel S_i, and let (u_k, v_k) represent the spatial coordinates (row and column) of the kth pixel. The unnormalized centroid coordinates $(\bar{u}_i, \bar{v}_i)$ are derived by averaging the positions of constituent pixels, expressed as

$$\bar{u}_i = \frac{1}{|\Omega_i|}\sum_{k\in\Omega_i} u_k, \quad \bar{v}_i = \frac{1}{|\Omega_i|}\sum_{k\in\Omega_i} v_k \tag{3}$$

where $|\Omega_i|$ is the number of pixels in S_i. To ensure scale invariance across different image resolutions, the final centroid vector $\mathbf{c}_i \in \mathbb{R}^2$ is obtained by normalizing these coordinates to the range $[0, 1]$:

$$\mathbf{c}_i = \left[\frac{\bar{u}_i}{H}, \frac{\bar{v}_i}{W}\right]^T \tag{4}$$

where H and W denote the height and width of the input image, respectively.

Based on the set of centroids $\mathcal{C} = \{\mathbf{c}_1, \ldots, \mathbf{c}_N\}$, we construct a sparse centroid-based neighbor matrix, denoted as $\mathbf{A}_{gsp} \in \mathbb{R}^{N\times N}$. To effectively encode local geometric structures, we employ a spatial **Top-K** strategy. For each centroid $\mathbf{c}_i$, we identify the set of its K nearest spatial neighbors, denoted as $\mathcal{N}_i^{spa}$, based on the Euclidean distance. The elements of $\mathbf{A}_{gsp}$ are then defined as

$$\mathbf{A}_{gsp}^{(i,j)} = \begin{cases} \exp\left(-\dfrac{\|\mathbf{c}_i - \mathbf{c}_j\|_2^2}{2\sigma^2}\right), & \text{if } j \in \mathcal{N}_i^{spa} \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where σ controls the distance sensitivity.

Crucially, $\mathbf{A}_{gsp}$ acts as a geometric structural prior rather than a static mask. It defines a spatially plausible candidate neighborhood for each node, effectively constraining the subsequent attention mechanism. This ensures that the dynamic graph learns semantic affinities within a geometrically guided search space, filtering out spurious long-range connections while focusing on semantically consistent regions.

B. Feature Embedding and the Multiscale CNN Branch

The proposed framework begins with a shared feature embedding module to regularize the raw input, followed by a dedicated convolutional branch designed to capture pixel-level spatial patterns. This architecture ensures that fine-grained spatial details are preserved to complement the superpixel-based graph learning.

1) Pixel-Wise Spectral Embedding (Stem): Raw HSIs are characterized by high-dimensional spectral bands, which often lead to information redundancy and computational inefficiency. To address this, we employ a Spectral Stem module as the global feature entry point, as illustrated in the architecture diagram Fig. 1. This module aims to project the raw spectral signatures into a compact latent space before any spatial branching occurs.

Specifically, a 1×1 convolutional layer is utilized to reduce the spectral dimensionality while preserving the original spatial resolution. By performing pixel-wise linear combinations across bands, this layer effectively compresses redundant information. To accelerate network convergence and prevent overfitting, batch normalization (BN) is applied immediately after convolution to standardize the feature distribution. The nonlinear mapping is subsequently introduced via the

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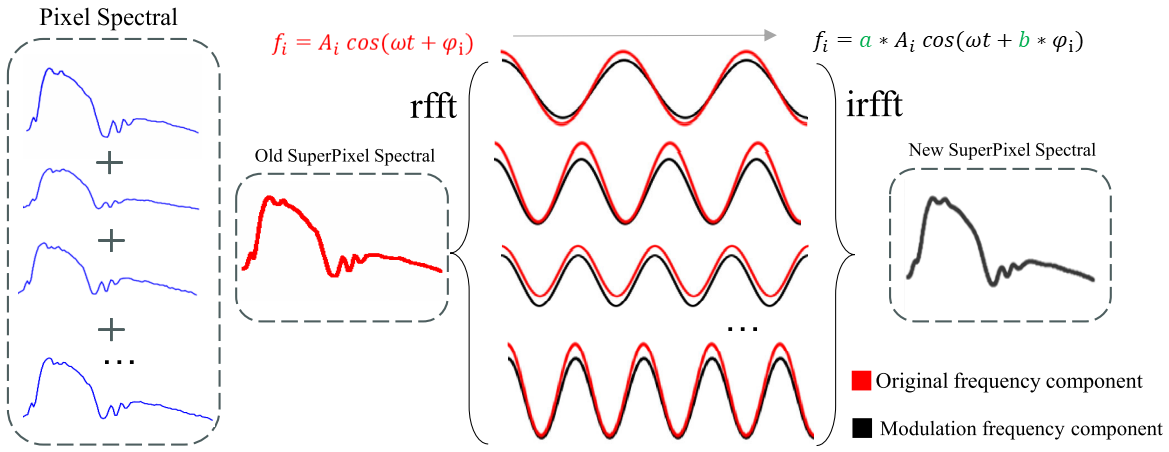


Fig. 2. Overview of the proposed FDSM for spectral refinement in the Fourier domain.

LeakyReLU activation function. Let $\mathbf{X}_{raw} \in \mathbb{R}^{H\times W\times B}$ denote the input image. The embedding process is formulated as

$$\mathbf{H}_{emb} = \sigma\left(\mathcal{B}\left(\mathbf{W}_{1\times 1} * \mathbf{X}_{raw} + \mathbf{b}\right)\right) \tag{6}$$

where $\mathbf{W}_{1\times 1}$ represents the learnable weights, $\mathcal{B}(\cdot)$ denotes batch normalization, and $\sigma(\cdot)$ is the activation function. The resulting feature map $\mathbf{H}_{emb}$ serves as the robust initialization for both the subsequent graph construction and the multiscale convolutional branch.

2) Multiscale CNN Branch: While the parallel graph branch excels at modeling long-range semantic dependencies via superpixel aggregation, it operates at a coarser spatial granularity and omits explicit pixel-level detail. To explicitly model multiscale spatial context at the pixel level, we design a Multiscale CNN branch that operates directly on the pixel grid features $\mathbf{H}_{emb}$.

To balance feature extraction capability with model complexity, we adopt depthwise separable convolutions (DSC) instead of standard convolutions, as they significantly reduce the parameter count by decoupling the operation into depth-wise spatial filtering and point-wise channel mixing. We employ a hierarchical cascading strategy to capture spatial features at varying scales, where the embedding is first processed by a 3×3 DSC block to capture detailed local textures and boundary information, followed by a 7×7 DSC block that expands the receptive field to aggregate broader contextual information.

In each block, BN and LeakyReLU are integrated to stabilize gradient propagation. The output feature at spatial position (i, j) of the lth depth-wise layer can be mathematically expressed as

$$\mathbf{Z}^l_{i,j} = \sigma\left(\sum_{u=-k'}^{k'}\sum_{v=-k'}^{k'} \mathbf{W}^l_{u,v}\cdot \mathbf{H}^{l-1}_{i+u,j+v} + \mathbf{b}^l\right) \tag{7}$$

where $\mathbf{H}^{l-1}$ and $\mathbf{Z}^l$ denote the input and output features of the layer, respectively. k' determines the kernel radius (e.g., $k' = 1$ for the 3×3 stage and $k' = 3$ for the 7×7 stage). This progressive design ensures that the network captures a comprehensive range of pixel-level spatial features, providing a crucial local complement to the region-based representations from the graph branch.

C. Fourier-Domain Spectral Modulation

Although superpixel aggregation effectively reduces spatial redundancy, the resulting node spectral mix semantic information with high-frequency artifacts such as boundary blending and noise. In the original spectral domain, these components are spectrally overlapped and difficult to separate. In contrast, they occupy distinct frequency bands: semantic content is concentrated in low frequencies, while aliasing artifacts appear primarily in high frequencies. To exploit this separability, we introduce the FDSM module. As illustrated in Fig. 2, FDSM transforms superpixel spectra into the Fourier domain, applies a learnable complex-valued gating mechanism to independently modulate the amplitude and phase of each frequency component, and reconstructs a refined node representation via inverse Fourier transform.

1) DFT and Complex Decomposition: Let $\mathbf{x} \in \mathbb{R}^C$ denote the spectral feature vector of a superpixel node, where $x[n]$ represents the intensity at the nth band. We first transform the signal into the frequency domain using the discrete Fourier transform (DFT)

$$X[k] = \sum_{n=0}^{C-1} x[n]\cdot e^{-j\frac{2\pi}{C}kn}, \quad 0 \le k < K \tag{8}$$

where $K = \lfloor C/2\rfloor + 1$. Since the input $\mathbf{x}$ is real-valued, we utilize the real-valued FFT (rFFT) to reduce computational redundancy. To explicitly derive the complex representation used in our modulation, we invoke **Euler's formula** ($e^{-j\theta} = \cos\theta - j\sin\theta$). The DFT formula can thus be expanded as

$$\begin{aligned} X[k] &= \sum_{n=0}^{C-1} x[n]\left[\cos\left(\frac{2\pi}{C}kn\right) - j\sin\left(\frac{2\pi}{C}kn\right)\right] \\ &= \underbrace{\sum_{n=0}^{C-1} x[n]\cos\left(\frac{2\pi}{C}kn\right)}_{u[k]} + j\underbrace{\sum_{n=0}^{C-1} -x[n]\sin\left(\frac{2\pi}{C}kn\right)}_{v[k]}. \end{aligned} \tag{9}$$

Through this decomposition, the frequency component $X[k]$ is represented in its canonical complex form $X[k] = u[k] + jv[k]$, where $u[k], v[k] \in \mathbb{R}$.

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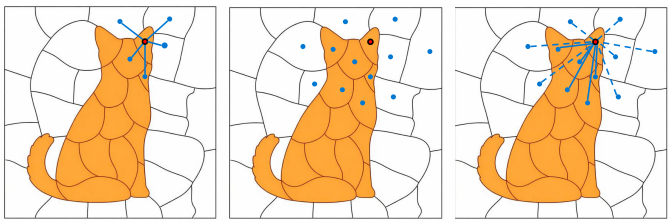


Fig. 3. Comparison of graph construction strategies. (a) Fixed local adjacency based on spatial proximity. (b) Spatial prior candidates defined by centroid distances. (c) Content-aware refinement, where solid and dashed lines represent learned semantic connections and suppressed noise edges, respectively.

2) Learnable Complex Frequency Modulation: To suppress high-frequency artifacts and calibrate discriminative spectral signatures, we introduce a learnable complex weight vector $\mathbf{w} \in \mathbb{C}^K$. By parameterizing $\mathbf{w}$ through its real and imaginary components as $\mathbf{w}[k] = a[k] + jb[k]$ (where $a[k], b[k] \in \mathbb{R}$), the module facilitates joint optimization of spectral content in the Fourier domain. The spectral modulation is executed via element-wise complex multiplication $\widetilde{X}[k] = X[k] \cdot \mathbf{w}[k]$. To this end, the operation is formulated as

$$\begin{aligned}\widetilde{X}[k] &= (u[k] + jv[k]) \cdot (a[k] + jb[k]) \\ &= (u[k]a[k] - v[k]b[k]) + j(u[k]b[k] + v[k]a[k]).\end{aligned} \tag{10}$$

This mechanism enables the network to jointly regulate the amplitude and phase of each frequency component in an end-to-end manner through the coupled interaction between the real and imaginary parts. Such complex-valued modulation endows the model with richer representational capacity in the Fourier domain, allowing it to capture more flexible spectral transformations and better characterize subtle yet discriminative frequency patterns, while effectively suppressing the high-frequency noise introduced by superpixel abstraction.

3) Inverse Reconstruction: Finally, the modulated frequency features are mapped back to the spectral domain using the inverse DFT (IDFT)

$$\tilde{x}[n] = \frac{1}{C} \sum_{k=0}^{C-1} \widetilde{X}[k] \cdot e^{j\frac{2\pi}{C}kn}. \tag{11}$$

The resulting $\tilde{x}[n]$ represents the refined spectral signature, which retains the original semantic structure while suppressing high-frequency fluctuations and noise artifacts.

These refined superpixel features are then used for the subsequent dynamic graph construction. Since the adaptive edge weights are computed from the similarities among node features, the FDSM-enhanced representations provide more reliable semantic cues for edge reweighting and neighbor selection, thereby promoting a more discriminative and stable dynamic graph topology.

D. DSSG Learning

Given the refined superpixel node features $\mathbf{X}$ (from FDSM) and the geometric spatial prior $\mathbf{A}_{gsp}$, our goal is to learn a task-adaptive graph structure that balances semantic relevance and spatial plausibility. Fig. 3 illustrates the key insight: dynamic graph construction can be viewed as a *search space optimization problem*. Unconstrained self-attention over all node pairs is computationally expensive and prone to spurious long-range connections, while rigid geometric adjacency restricts modeling flexibility by ignoring semantic homogeneity across nonadjacent regions. To address this, we propose a dynamic spectral–spatial graph (DSSG) mechanism that treats $\mathbf{A}_{gsp}$ as a structural before defining a candidate neighborhood for each node. Within this spatially plausible region, spectral similarity is used to dynamically weight edges via self-attention. The self-attention weights are computed only over the candidate neighborhood defined by $\mathbf{A}_{gsp}$, which eliminates irrelevant connections and focuses semantic modeling on spatially coherent regions. The resulting dynamic adjacency matrix $\mathbf{A}_{dyn}$ is generated end-to-end by attending only to nodes permitted by $\mathbf{A}_{gsp}$, thus integrating spatial constraints as an inductive bias rather than a post hoc fusion.

1) Positional Encoding and Feature Projection: Although superpixel aggregation reduces data redundancy, flattening superpixel nodes into a 1-D sequence for GNN input removes their original ordering information. Since the subsequent graph construction is purely content-driven, we do not aim to encode precise 2-D spatial coordinates here, but instead introduce Sinusoidal Position Encoding based on sequence indices to provide a lightweight ordering cue for the following attention-based feature projection. This 1-D nonparametric design preserves the content-oriented nature of graph construction while avoiding extra parameters and unnecessary geometric bias. Let *pos* denote the node index in the sequence and d_k be the embedding dimension. The positional encoding $\mathbf{P} \in \mathbb{R}^{N \times d_k}$ is defined as

$$\begin{aligned}\mathbf{P}(pos, 2j) &= \sin\left(pos/10000^{2j/d_k}\right) \\ \mathbf{P}(pos, 2j+1) &= \cos\left(pos/10000^{2j/d_k}\right).\end{aligned} \tag{12}$$

The node features $\mathbf{X}$ are then projected into the query and key spaces, with positional signals integrated to guide the subsequent geometric-aware attention

$$\mathbf{Q}_a = \mathbf{X}\mathbf{W}_Q + \mathbf{P}, \quad \mathbf{K}_a = \mathbf{X}\mathbf{W}_K + \mathbf{P} \tag{13}$$

where $\mathbf{W}_Q, \mathbf{W}_K$ are learnable projection matrices.

2) Geometrically Constrained Graph Construction: To strictly confine the spectral affinity learning within geologically plausible regions, we propose a fused attention mechanism modulated by a *geometric log-barrier*. Instead of performing an unconstrained global search, the centroid prior $\mathbf{A}_{gsp}$ is injected directly into the attention logits as a structural filter. By incorporating this geometric constraint, the network is forced to prioritize spectral correlations within a spatially local context, effectively preventing the propagation of noise from distant, unrelated superpixels. The unnormalized affinity matrix $\mathbf{E} \in \mathbb{R}^{N \times N}$ is formulated as

$$\mathbf{E} = \frac{\mathbf{Q}_a \mathbf{K}_a^\top}{\sqrt{d_k} \cdot \tau} + \frac{\ln\left(\mathbf{A}_{gsp}\right)}{\tau} \tag{14}$$

where τ is the temperature scaling factor and $\ln(\mathbf{A}_{gsp})$ acts as a structural bias. For nonadjacent node pairs where $\mathbf{A}_{gsp}^{(i,j)} = 0$, the logarithmic penalty forces the affinity toward $-\infty$, thereby nullifying the connection regardless of spectral similarity.

This mechanism effectively prunes the search space, ensuring that spectral dependencies are refined *only* within

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the predefined geometric neighborhood. As illustrated in Fig. 3(b), such structural regularization maintains the topological integrity of the evolving graph, allowing the model to capture discriminative features even with limited labeled samples.

The calculated affinity $\mathbf{E}$ yields a dense probability matrix $\mathbf{A}^{dense} = \text{Softmax}(\mathbf{E})$. However, directly using a dense attention graph would retain many weak or noisy connections, which may dilute the most relevant semantic relationships and increase computational burden in subsequent graph propagation. To suppress potential noise, enhance computational efficiency, and focus message passing on the most informative neighbors, a dynamic sparsification strategy is employed. Specifically, for each node i, the connectivity is restricted to its top-k nearest neighbors $\mathcal{N}_i$ in the feature space. The resulting sparse adjacency matrix $\mathbf{A}^{sparse}$ is defined as

$$\mathbf{A}_{ij}^{sparse} = \begin{cases} \mathbf{A}_{ij}^{dense}, & \text{if } j \in \mathcal{N}_i \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

To maintain a valid undirected graph topology, a symmetric structure is enforced via

$$\mathbf{A}_{dyn} = \max\left(\mathbf{A}^{sparse}, (\mathbf{A}^{sparse})^{\top}\right). \tag{16}$$

After row-normalization, $\mathbf{A}_{dyn}$ serves as the final refined topology for the subsequent GAT layers.

3) Multihead Graph Attention Propagation: With the learned dynamic topology $\mathbf{A}_{dyn}$, we employ a GAT to aggregate features. The attention coefficient e_{ij} between node i and its neighbor $j \in \mathcal{N}_i$ (where $\mathcal{N}_i = \{j \mid \mathbf{A}_{dyn}^{(i,j)} > 0\}$) is computed via a single-layer feedforward network

$$e_{ij} = \text{LeakyReLU}\left(\mathbf{a}^{\top}\left[\mathbf{W}_g\mathbf{h}_i \,\|\, \mathbf{W}_g\mathbf{h}_j\right]\right) \tag{17}$$

where $\mathbf{W}_g$ is the weight matrix for the GAT layer, $\mathbf{a}$ is the attention vector, and $\|$ denotes concatenation. The coefficients are normalized across the neighborhood using the Softmax function

$$\alpha_{ij} = \frac{\exp\left(e_{ij}\right)}{\sum_{k\in\mathcal{N}_i}\exp\left(e_{ik}\right)}. \tag{18}$$

Finally, we update the node embeddings by aggregating the neighbor features. To stabilize the learning process, we utilize a multihead attention mechanism with M heads, concatenating their outputs

$$\mathbf{h}_i' = \Big\Vert_{m=1}^{M} \sigma\left(\sum_{j\in\mathcal{N}_i}\alpha_{ij}^{(m)}\mathbf{W}_g^{(m)}\mathbf{h}_j\right). \tag{19}$$

Through this multihead mechanism, node representations are iteratively refined by aggregating spectral features from semantically related neighbors, a process explicitly regularized by the spatial centroid constraints. Let $\mathbf{H}^{(l)} = [\mathbf{h}_1^{(l)}, \ldots, \mathbf{h}_N^{(l)}]^{\top} \in \mathbb{R}^{N\times F}$ represent the node feature matrix at layer l. By stacking L such GAT layers, the model extracts the high-level node embedding matrix $\mathbf{H}^{(L)}$, which captures complex latent dependencies and serves as the input for the subsequent decoder.

4) Superpixel-to-Pixel Decoder: Since graph reasoning is performed on N superpixel nodes, we employ a *superpixel-to-pixel decoder* to recover dense grid features. With the superpixel association matrix $\mathbf{Q} \in \mathbb{R}^{HW\times N}$ defined in (1), the refined node embeddings $\mathbf{H}^{(L)} \in \mathbb{R}^{N\times F}$ are decoded to the pixel space as

$$\mathbf{Z}_{graph} = \text{Decoder}\left(\mathbf{H}^{(L)}; \mathbf{Q}\right) \tag{20}$$

$$= \text{Reshape}\left(\mathbf{Q}\mathbf{H}^{(L)}\right) \in \mathbb{R}^{H\times W\times F} \tag{21}$$

where Reshape$(\cdot)$ restores the spatial layout, and each pixel inherits the embedding of its associated superpixel node.

E. Weighted Feature Fusion and Classification

The two branches operate on fundamentally different domains: the CNN branch processes the pixel grid to preserve high-frequency spatial details (e.g., edges and textures), while the dynamic graph branch models a non-Euclidean superpixel graph to capture low-frequency semantic consistency. To reconcile these heterogeneous yet complementary representations, we align them in a shared feature space and fuse them via a learnable weight λ. The final fused representation $\mathbf{Z} \in \mathbb{R}^{H\times W\times C}$ is computed as

$$\mathbf{Z} = \lambda \cdot \mathbf{Z}_{graph} + (1-\lambda)\cdot \mathbf{Z}_{conv} \tag{22}$$

where $\mathbf{Z}_{graph}$ and $\mathbf{Z}_{conv}$ denote the global features projected from the graph branch and the local features from the convolutional branch, respectively.

The network is optimized using the cross-entropy loss function. Given the scarcity of labeled data in HSI scenarios, the loss is exclusively computed over the indices within the labeled training set $\mathcal{D}_{train}$

$$\mathcal{L} = -\frac{1}{N}\sum_{i\in\mathcal{D}_{train}}\sum_{c=1}^{C} y_{i,c}\log\left(p_{i,c}\right) \tag{23}$$

where N is the total number of training samples. Here, $y_{i,c}$ denotes the one-hot ground-truth label for class c, and $p_{i,c}$ represents the predicted probability of pixel i belonging to class c, as determined by the final Softmax layer.

III. Experimental Results

A. Data Description

To comprehensively evaluate the performance of the proposed model, we conduct experiments on four hyperspectral datasets: the IP dataset, the PU dataset, the Salinas dataset, and the UP dataset.

1) *IP:* Acquired by the AVIRIS sensor over an agricultural test site, with 145×145 pixels and 220 spectral bands. After removing low signal-to-noise and water-vapor absorption bands, 200 bands and 10 366 labeled pixels from 16 land-cover classes are used.
2) *PU:* Collected by the ROSIS sensor over an urban area, with 610×340 pixels and 115 bands. After discarding noisy bands, 103 bands remain, providing 42 776 labeled pixels from nine classes.
3) *Salinas (SA):* Acquired over the Salinas Valley, with 512×217 pixels and 224 bands. After removing water-absorption and noisy bands, 204 bands and 54 129 labeled pixels from 16 crop classes are retained.

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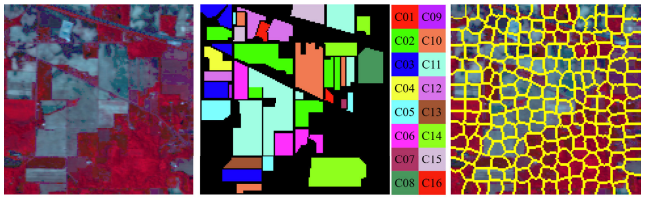


Fig. 4. IP dataset. (a) False-color image. (b) Ground-truth map. (c) Superpixel segmentation map.

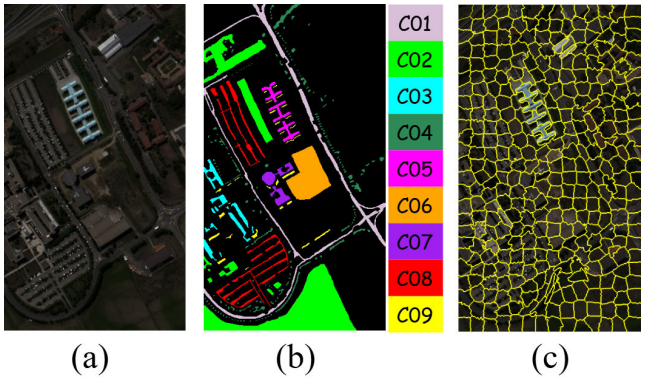


Fig. 5. PU dataset. (a) False-color image. (b) Ground-truth map. (c) Superpixel segmentation map.

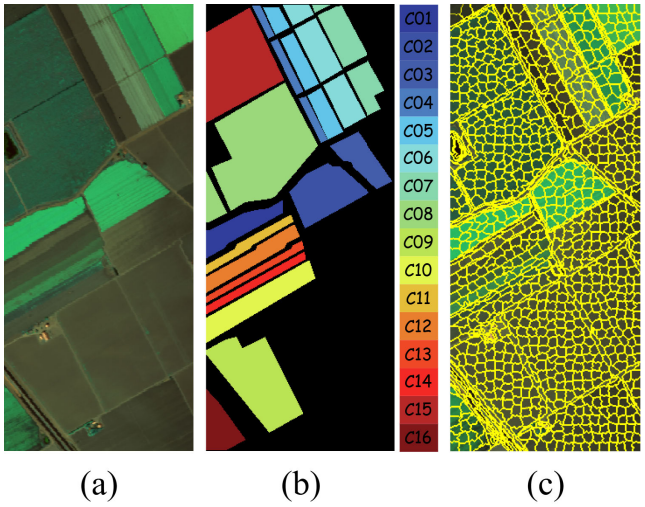


Fig. 6. SA dataset. (a) False-color image. (b) Ground-truth map. (c) Superpixel segmentation map.

4) *UP:* This dataset covers the landing site of the Zhurong rover (Tianwen-1 mission), captured by the CRISM instrument. The image size is 478×595 pixels, with 432 spectral bands retained after removing water-absorption and noisy channels.

The false-color composites and corresponding ground-truth maps of these datasets are shown in Figs. 4–7, while the class names and the numbers of training and test samples are listed in Tables I–IV. Specifically, ten labeled samples per class are randomly selected for training, and the remaining labeled samples are used for testing.

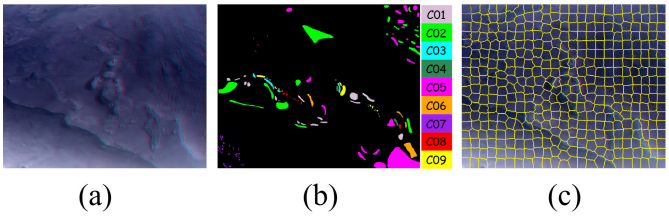


Fig. 7. UP dataset. (a) False-color image. (b) Ground-truth map. (c) Superpixel segmentation map.

TABLE I
CATEGORY NAMES AND NUMBER OF SAMPLES DIVIDED IN THE IP DATASET

Class ID	Class Name	Train	Test	Total
1	Alfalfa	10	36	46
2	Corn-notill	10	1418	1428
3	Corn-mintill	10	820	830
4	Corn	10	227	237
5	Grass-pasture	10	473	483
6	Grass-trees	10	720	730
7	Grass-pasture-mowed	10	18	28
8	Hay-windrowed	10	468	478
9	Oats	10	10	20
10	Soybean-notill	10	962	972
11	Soybean-mintill	10	2445	2455
12	Soybean-clean	10	583	593
13	Wheat	10	195	205
14	Woods	10	1255	1265
15	Buildings-Grass-Trees-Drives	10	376	386
16	Stone-Steel-Towers	10	83	93
	Total	160	10089	10249

TABLE II
CATEGORY NAMES AND NUMBER OF SAMPLES DIVIDED IN THE PU DATASET

Class ID	Class Name	Train	Test	Total
1	Asphalt	10	6621	6631
2	Meadows	10	18639	18649
3	Gravel	10	2089	2099
4	Trees	10	3054	3064
5	Painted metal sheets	10	1335	1345
6	Bare Soil	10	5019	5029
7	Bitumen	10	1320	1330
8	Self-Blocking Bricks	10	3672	3682
9	Shadows	10	937	947
	Total	90	42686	42776

1) Evaluation Setup: To quantitatively evaluate the classification performance, we employ three standard metrics: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (Kappa).

2) Environmental Setup: All experiments were implemented using PyTorch on a Linux workstation equipped with an NVIDIA GeForce RTX 3090 GPU. To ensure statistical reliability, each experiment was repeated ten times, and the results are reported as the mean $\pm$ standard deviation.

3) Parameter Setup: To ensure reproducibility and maximize the performance of GSDG-Net, we implemented a rigorous experimental configuration. All models were optimized using the Adam optimizer with a learning rate of 0.001

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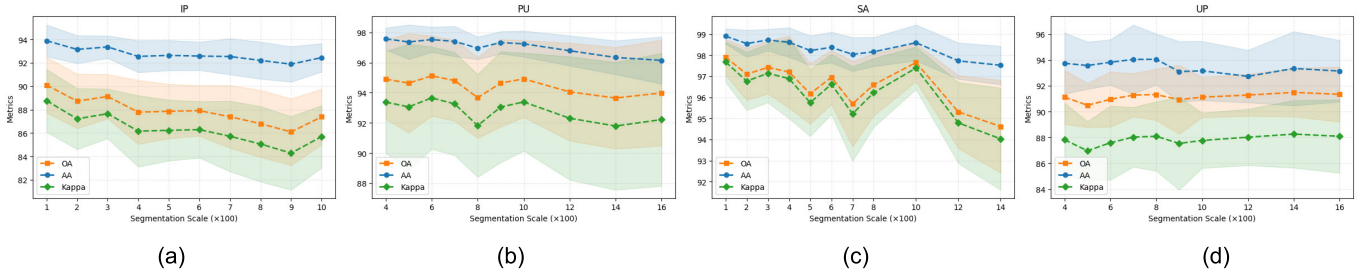


Fig. 8. Sensitivity analysis of superpixel scale (S). The results illustrate the classification performance (OA, AA, and Kappa) across four datasets with varying segmentation scales. (a) IP dataset. (b) PU dataset. (c) SA dataset. (d) UP dataset.

TABLE III
CATEGORY NAMES AND NUMBER OF SAMPLES DIVIDED IN THE SA DATASET

Class ID	Class Name	Train	Test	Total
1	Brocoli_green_weeds_1	10	1999	2009
2	Brocoli_green_weeds_2	10	3716	3726
3	Fallow	10	1966	1976
4	Fallow_rough_plow	10	1384	1394
5	Fallow_smooth	10	2668	2678
6	Stubble	10	3949	3959
7	Celery	10	3569	3579
8	Grapes_untrained	10	11261	11271
9	Soil_Vineyard_develop	10	6193	6203
10	Corn_senesced_green_weeds	10	3268	3278
11	Lettuce_romaine_4wk	10	1058	1068
12	Lettuce_romaine_5wk	10	1917	1927
13	Lettuce_romaine_6wk	10	906	916
14	Lettuce_romaine_7wk	10	1060	1070
15	Vineyard_untrained	10	7258	7268
16	Vineyard_vertical_trellis	10	1797	1807
	Total	160	53969	54129

TABLE IV
CATEGORY NAMES AND NUMBER OF SAMPLES DIVIDED IN THE UP DATASET

Class ID	**Class Name**	**Train**	**Test**	**Total**
1	Analcime	10	1741	1751
2	Bassanite	10	5797	5807
3	High-Ca Pyroxene	10	422	432
4	Illite/Muscovite	10	262	272
5	Low-Ca Pyroxene	10	6764	6774
6	Mg-Smectite	10	1156	1166
7	Monohydrated sulfate	10	330	340
8	Plagioclase	10	192	202
9	Prehnite	10	584	594
	Total	90	17248	17338

and weight decay of 1×10^{-4} over 400 epochs to facilitate stable convergence across all datasets.

B. Experimental Setting

1) Comparison Methods: To comprehensively evaluate the performance of the proposed GSDG-Net, we compare it against eight state-of-the-art methods. These baselines were carefully selected to cover representative CNN-based architectures (e.g., SSRN [25]) and recent advanced graph- and Transformer-based models (such as CEGCN [44], WFCG [52], DGFNet [53], GTFN [54], PSFormer [42], HGCN-MHF [48], GTCFN [55], and NESSGGCN [49]).

2) Sensitivity Analysis of Key Hyperparameters: Beyond basic optimization, we performed a comprehensive sensitivity analysis for four task-critical hyperparameters. First, we investigated the **superpixel scale** (S), which dictates the granularity of graph nodes. As illustrated in Fig. 8, the optimal scale reflects the spatial complexity of each scene: IP and SA achieve peak OA at $S = 100$, while the more intricate PU and UP datasets require $S = 600$ and $S = 800$, respectively. Regarding the geometric prior construction, we established the **spatial centroid sparsity** by limiting each node to its 15 nearest spatial neighbors based on Euclidean distance between centroids. The value was selected empirically, as it provides a good tradeoff between sufficient local geometric coverage and limited redundancy in the candidate neighborhood graph.

Furthermore, we examined the **fusion coefficient** (λ) to identify the optimal balance between regional semantics and pixel-level details. The findings presented in Fig. 9 reveal that the IP and SA datasets achieve peak OA at $\lambda = 0.95$, suggesting that global structural information is dominant. In contrast, the accuracy for PU and UP peaks at $\lambda = 0.6$, indicating a more balanced requirement for global consistency and local textures.

Finally, we explored the hyperparameter space for the dynamic graph construction, specifically the **embedding dimension** (d_k) and the **sparsity level** (k). We tested $d_k \in \{8, 16, \ldots, 48\}$ and $k \in \{4, 6, \ldots, 16\}$, with the results visualized in Fig. 10. The OA exhibits a "mountain-peak" trend, where $d_k = 16$ and $k = 8$ yield optimal results. While lower k values fail to capture sufficient semantic neighbors, excessive k or d_k introduces topological noise and unnecessary computational overhead.

In addition to the individual sensitivity trends shown in Figs. 8–Figs. 10, these hyperparameters also exhibit certain coupling effects. Specifically, the superpixel scale S and the fusion coefficient λ jointly affect the balance between regional semantic abstraction and pixel-level detail preservation. Datasets with relatively simpler spatial layouts, such as IP and SA, tend to benefit from smaller S together with larger λ, indicating a stronger reliance on graph-level structural modeling. In contrast, PU and UP favor larger segmentation scales and more moderate fusion weights, suggesting that a better tradeoff between global consistency and local texture modeling is required. Meanwhile, the embedding dimension d_k and sparsity level k jointly influence the quality of dynamic

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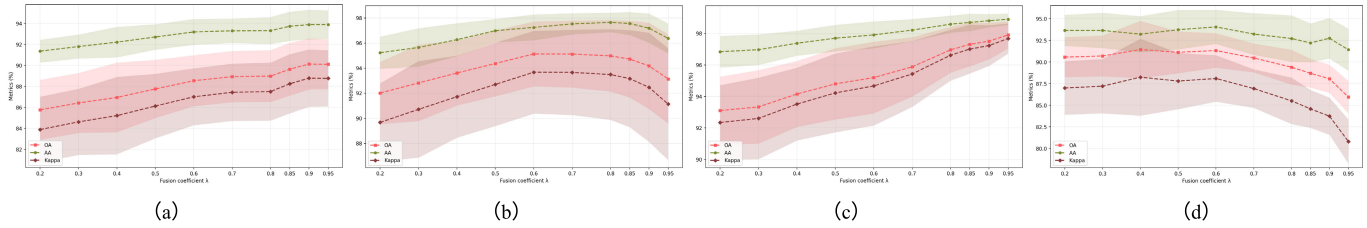


Fig. 9. Sensitivity analysis of the fusion coefficient λ. The shaded areas represent the standard deviation, indicating the stability of the model. (a) IP dataset. (b) PU dataset. (c) SA dataset. (d) UP dataset.

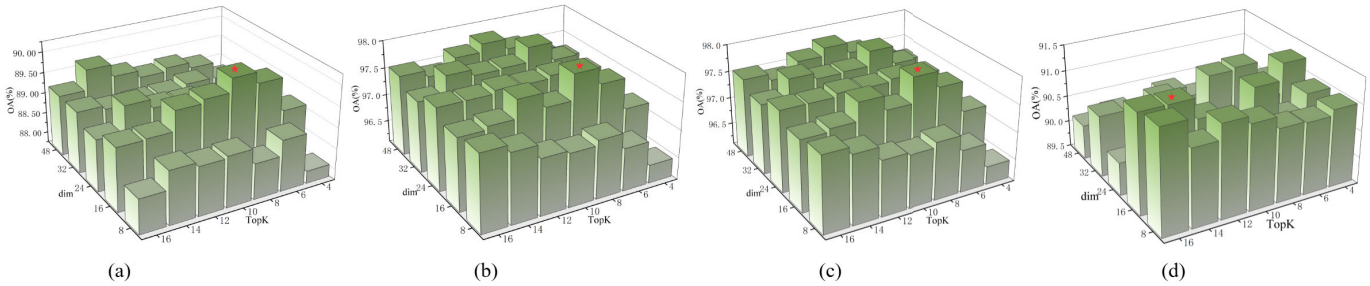


Fig. 10. Parameter analysis of the dynamic graph construction module. The 3-D bar charts visualize the impact of embedding dimension d_k and Top-k sparsity on OA. (a) IP. (b) Salinas. (c) PaviaU. (d) Utopia.

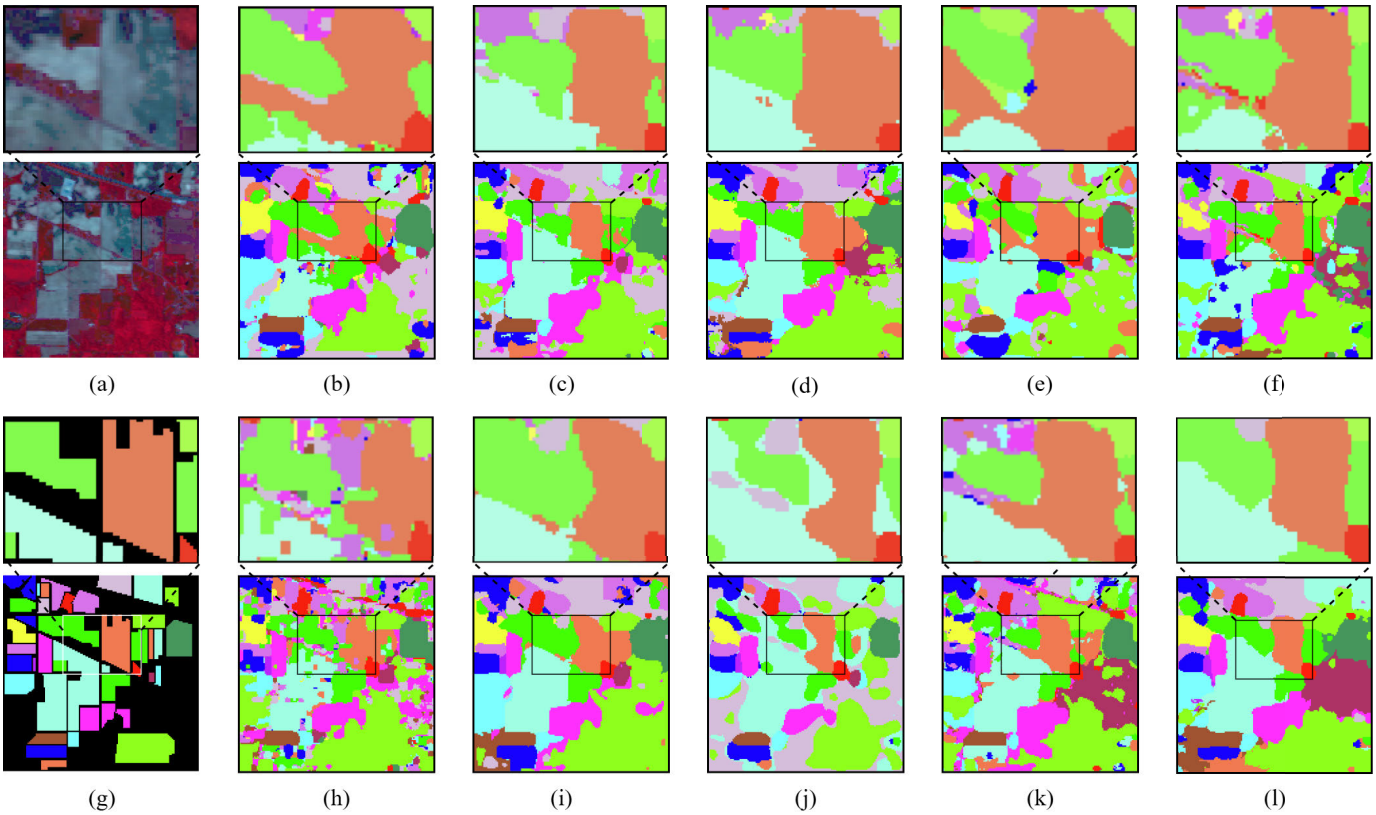


Fig. 11. Classification maps of different methods on the IP dataset. (a) False-color image. (b) SSRN. (c) CEGCN. (d) WFCG. (e) DGFNet. (f) GTFN. (g) Ground truth. (h) PSFormer. (i) HGCN-MHF. (j). GTCFN. (k) NESSGGCN. (l) Ours.

graph construction. A too small d_k limits relation modeling, whereas an excessively large k tends to introduce redundant or noisy neighbors; therefore, only a proper combination of the two can achieve the best tradeoff between expressive capacity and robustness. These observations indicate that the performance of GSDG-Net depends on a coordinated balance among node granularity, branch fusion, and graph connectivity, rather than on the isolated tuning of a single parameter. Based

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on the above analysis, these parameter settings were adopted in all comparative experiments.

C. Classification Results and Analysis

The quantitative classification performance on four benchmark datasets is summarized in Tables V–VIII. Based on the reported results, we provide the following analysis.

1) IP Dataset: The IP dataset is primarily challenged by low spatial resolution and severe mixed-pixel effects, leading to pixel confusion and detail loss. As indicated in Table V, GSDG-Net achieves the highest OA of 90.10%, outperforming the second-best method (HGCN-MHF) by 2.48%. While Transformer-based methods like PSFormer struggle to maintain consistency in such noisy environments, GSDG-Net effectively suppresses high-frequency aliasing artifacts through Fourier-domain modulation. As shown in the classification map in Fig. 11, our method yields evidently smoother regions and sharper boundaries compared to the SSRN and GTFN, further validating its robust feature extraction capability in low-resolution HSI scenes.

2) PU Dataset: Although the PU dataset has high spatial resolution, the irregular distribution of urban objects makes it difficult to model consistent structural dependencies. According to Table VI, GSDG-Net achieves a significant OA of 95.12%, representing an improvement of 1.50% to 40.90% over the compared methods. Specifically, GSDG-Net outperforms the representative CNN method (SSRN) by a wide margin, which underscores the necessity of synergizing local textures with global structural cues. By resolving spectral ambiguities that often plague static graph models like CEGCN, our framework ensures that both intricate urban details and large-scale structures are accurately represented, leading to higher regional consistency.

3) SA Dataset: In the SA dataset, the regular spatial distribution typically results in higher OA; however, the bottleneck lies in distinguishing subtle spectral discrepancies between similar vegetation varieties. Table VII shows that GSDG-Net reaches a state-of-the-art OA of 97.91%, surpassing the latest graph-based NESSGGCN. Unlike static GNNs that may suffer from over-smoothing in large agricultural fields, GSDG-Net adaptively refines the graph topology based on semantic affinity. This ensures that the message-passing process remains discriminative for fine-grained classification, effectively maintaining high intraclass homogeneity without sacrificing boundary details.

4) UP Dataset: The UP dataset presents extreme spectral variability and complex mineral topography, demanding high topological robustness for graph construction. According to Table VIII, GSDG-Net maintains a robust lead with an OA of 91.32%, outperforming the second-best result by nearly 1.40%. While many advanced methods suffer from performance fluctuations on this intricate Martian landscape, GSDG-Net captures more reliable mineral signatures by imposing strict geometric constraints on the evolving graph. This mechanism effectively prevents the formation of spurious connections across disjoint geological units, demonstrating the superior stability and generalization of our method on nonterrestrial datasets.

IV. Discussion

A. Ablation Study

1) Ablation Experiments on the FDSM and DSSG Modules: To verify the effectiveness of our core contributions, we evaluate the incremental benefits of FDSM and DSSG in Table IX, together with their parameter counts. As shown, adding either module consistently improves the classification accuracy over the baseline where both modules are removed, while introducing only a small increase in model complexity.

The FDSM module acts as a learnable spectral filter to suppress high-frequency noise and ambiguity in superpixel features. By transforming node features into the frequency domain, FDSM adaptively emphasizes discriminative spectral components. As shown in Table IX, incorporating FDSM alone improves the OA on the IP and PU datasets by 0.22% and 0.34%, respectively, with only a slight increase in parameters.

The DSSG module addresses the limitation of topological rigidity. When DSSG is disabled ($\times$ in Table IX), the model relies on a **fixed local adjacency** based strictly on spatial proximity, as illustrated in Fig. 3(a). By introducing dynamic graph refinement, the model can remove spatially adjacent but semantically irrelevant edges while establishing long-range semantic connections, corresponding to the content-aware refinement in Fig. 3(c). This change alone significantly improves the OA, for example, from 93.10% to 94.70% on the PU dataset, again with only limited parameter growth.

The full GSDG-Net achieves the best performance across all datasets while remaining parameter-efficient. For example, on the PU dataset, the OA increases from 93.10% to 95.12%, whereas the parameter count increases only from 106.537K to 110.665K. These results demonstrate that both FDSM and DSSG provide clear performance gains with modest additional parameters.

2) Comparison of FDSM With Alternative Spectral Modeling Strategies: To further validate the effectiveness of the proposed FDSM module, we compared it with several related spectral modeling strategies, including 1-D convolution, band attention, and real-valued frequency gating, under the same backbone and training settings. As shown in Table X, FDSM achieves the best OA on all four datasets and the best AA on most datasets, while maintaining a parameter count comparable to or lower than the competing methods. In particular, compared with real-valued frequency gating, FDSM yields consistently better results, suggesting that jointly modeling the real and imaginary components in the Fourier domain provides a more expressive way to capture spectral frequency information. These results indicate that the proposed complex-valued Fourier modulation is more effective in suppressing intrasuperpixel spectral noise and enhancing discriminative spectral representations than conventional spectral modeling strategies.

3) Ablation Experiments on the Dynamic Graph Construction Mechanisms: We proposed a sophisticated graph generation strategy involving Top-k sparsity ($Topk$), position encoding (pos), and a spatial prior matrix (A_{gsp}). To validate their individual contributions, we conducted a leave-one-out experiment as summarized in Table XI. Crucially, the absence of the geometric spatial prior (A_{gsp}) in Case 3 leads

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TABLE V

CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE IP DATASET

No.	SSRN (2018)	CEGCN (2021)	WFCG (2022)	DGFNet (2023)	GTFN (2023)	PSFormer (2024)	HGCN-MHF (2024)	GTCFN (2025)	NESSGGCN (2025)	GSDG-Net (Ours)
1	99.44 ± 1.11	99.17 ± 1.27	99.17 ± 1.27	99.17 ± 1.27	**99.72 ± 0.83**	98.33 ± 3.33	**99.72 ± 0.83**	91.11 ± 4.08	53.11 ± 24.33	99.17 ± 1.27
2	62.06 ± 7.45	62.18 ± 10.70	75.74 ± 8.37	74.49 ± 10.82	72.19 ± 8.15	62.33 ± 17.49	76.94 ± 11.29	29.67 ± 8.16	76.89 ± 8.78	**79.58 ± 7.91**
3	71.15 ± 9.67	71.59 ± 14.31	78.11 ± 9.89	70.70 ± 10.82	72.23 ± 9.63	51.43 ± 10.69	**84.37 ± 8.91**	64.89 ± 19.23	79.21 ± 14.51	82.23 ± 10.44
4	95.95 ± 4.72	98.63 ± 1.48	**99.30 ± 0.82**	92.20 ± 8.18	95.29 ± 2.41	91.50 ± 8.59	98.68 ± 1.64	77.44 ± 4.87	60.30 ± 15.70	99.03 ± 1.56
5	**87.48 ± 6.10**	80.17 ± 9.13	82.49 ± 7.56	80.34 ± 9.78	79.96 ± 6.41	72.20 ± 9.33	86.08 ± 6.46	69.83 ± 8.84	75.96 ± 10.08	82.98 ± 8.24
6	96.15 ± 3.54	98.67 ± 1.16	98.93 ± 1.48	92.85 ± 4.78	95.17 ± 2.13	93.61 ± 6.31	97.86 ± 2.24	56.68 ± 14.27	**99.85 ± 0.23**	98.40 ± 1.44
7	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.44 ± 1.67	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	84.07 ± 27.32	**100.00 ± 0.00**
8	96.52 ± 4.23	**100.00 ± 0.00**	**100.00 ± 0.00**	94.47 ± 8.76	99.04 ± 0.87	97.48 ± 5.27	99.85 ± 0.22	94.42 ± 2.55	99.89 ± 0.19	99.98 ± 0.06
9	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	58.95 ± 15.03	**100.00 ± 0.00**
10	77.99 ± 14.41	88.65 ± 5.77	86.81 ± 6.75	70.76 ± 6.93	80.04 ± 5.91	74.46 ± 9.88	85.40 ± 5.34	82.27 ± 14.22	86.60 ± 6.76	**88.22 ± 6.07**
11	64.43 ± 4.65	79.99 ± 4.70	80.05 ± 7.38	60.91 ± 11.87	74.25 ± 4.71	66.92 ± 11.28	81.86 ± 10.66	45.46 ± 23.03	**92.98 ± 1.42**	88.73 ± 3.63
12	73.12 ± 13.16	81.10 ± 9.15	**93.60 ± 3.84**	76.91 ± 9.86	81.18 ± 4.93	69.71 ± 12.99	91.07 ± 5.40	60.62 ± 7.73	44.96 ± 17.61	92.23 ± 4.25
13	99.80 ± 0.25	**99.95 ± 0.15**	99.90 ± 0.21	98.15 ± 1.88	99.28 ± 0.73	98.62 ± 2.18	99.59 ± 0.85	94.15 ± 2.68	98.42 ± 1.41	**99.95 ± 0.15**
14	93.75 ± 5.15	96.79 ± 3.38	96.58 ± 4.17	85.00 ± 6.08	93.61 ± 4.29	94.85 ± 7.39	94.46 ± 6.46	68.13 ± 23.82	**98.29 ± 2.78**	97.33 ± 3.81
15	89.71 ± 7.75	95.05 ± 6.61	97.79 ± 3.04	86.17 ± 12.15	90.45 ± 6.44	84.95 ± 7.95	99.49 ± 0.42	67.69 ± 17.62	89.35 ± 7.94	**99.60 ± 0.45**
16	98.92 ± 1.90	**100.00 ± 0.00**	99.76 ± 0.72	94.94 ± 4.66	99.28 ± 0.96	99.16 ± 1.87	99.39 ± 1.47	**100.00 ± 0.00**	93.59 ± 11.53	**100.00 ± 0.00**
OA	77.76 ± 1.80	83.71 ± 2.54	86.92 ± 2.46	76.11 ± 2.68	82.00 ± 2.56	75.25 ± 3.04	87.62 ± 2.83	59.49 ± 7.96	85.86 ± 2.12	**90.10 ± 2.38**
AA	87.90 ± 1.53	90.18 ± 1.52	92.60 ± 1.16	86.07 ± 1.36	89.45 ± 1.32	84.72 ± 1.44	93.42 ± 1.53	75.15 ± 3.67	80.78 ± 3.58	**93.88 ± 1.36**
Kappa	74.95 ± 2.07	81.56 ± 2.85	85.19 ± 2.75	73.25 ± 2.87	79.68 ± 2.85	71.98 ± 3.26	85.97 ± 3.17	54.92 ± 8.14	83.83 ± 2.44	**88.76 ± 2.68**

TABLE VI

CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE PU DATASET

No.	SSRN (2018)	CEGCN (2021)	WFCG (2022)	DGFNet (2023)	GTFN (2023)	PSFormer (2024)	HGCN-MHF (2024)	GTCFN (2025)	NESSGGCN (2025)	GSDG-Net (Ours)
1	88.06 ± 8.37	92.44 ± 3.56	95.25 ± 3.00	66.21 ± 12.34	79.08 ± 5.35	88.98 ± 6.79	**98.34 ± 1.67**	85.23 ± 18.64	82.10 ± 9.96	94.99 ± 3.61
2	79.46 ± 7.77	89.54 ± 7.60	89.19 ± 4.65	37.20 ± 20.28	85.21 ± 5.37	76.81 ± 5.62	85.57 ± 7.79	74.86 ± 27.12	69.82 ± 12.58	**92.43 ± 5.82**
3	87.63 ± 4.83	97.31 ± 2.82	**97.78 ± 3.00**	56.96 ± 23.78	75.86 ± 7.12	90.11 ± 10.09	92.74 ± 14.05	68.02 ± 18.27	75.55 ± 10.72	96.03 ± 4.92
4	**96.22 ± 1.55**	94.09 ± 2.61	95.90 ± 2.06	65.38 ± 16.85	87.21 ± 7.21	94.86 ± 7.22	93.87 ± 2.08	47.29 ± 19.99	89.84 ± 4.41	95.72 ± 2.17
5	99.44 ± 0.99	99.85 ± 0.43	99.94 ± 0.12	84.11 ± 20.24	99.23 ± 0.77	**99.99 ± 0.04**	99.90 ± 0.13	99.63 ± 0.36	99.94 ± 0.10	99.75 ± 0.34
6	92.84 ± 7.36	98.54 ± 2.88	98.19 ± 2.63	65.44 ± 22.95	93.01 ± 3.01	89.37 ± 7.62	98.71 ± 2.91	77.96 ± 17.64	86.85 ± 9.67	**100.00 ± 0.00**
7	95.84 ± 2.98	99.71 ± 0.38	**100.00 ± 0.00**	81.71 ± 18.05	97.18 ± 3.61	99.36 ± 1.66	92.40 ± 22.02	73.48 ± 13.48	91.28 ± 7.90	**100.00 ± 0.00**
8	88.96 ± 5.83	95.99 ± 3.24	96.57 ± 3.41	65.82 ± 23.32	79.80 ± 5.93	89.88 ± 7.57	**98.54 ± 0.60**	69.34 ± 18.91	92.57 ± 3.93	96.75 ± 2.32
9	99.72 ± 0.35	99.54 ± 0.41	99.61 ± 0.53	78.93 ± 18.50	94.69 ± 3.86	**99.91 ± 0.08**	99.17 ± 0.66	89.14 ± 7.02	98.91 ± 1.59	99.58 ± 0.53
OA	86.36 ± 3.72	93.17 ± 3.79	93.62 ± 2.13	54.22 ± 12.42	85.41 ± 2.34	85.17 ± 3.02	92.11 ± 4.10	75.09 ± 9.99	79.64 ± 5.29	**95.12 ± 2.67**
AA	92.02 ± 1.41	96.82 ± 1.21	97.15 ± 0.90	66.86 ± 10.14	87.92 ± 1.20	92.14 ± 2.14	95.47 ± 3.78	76.10 ± 6.84	87.43 ± 2.12	**97.53 ± 0.84**
Kappa	82.65 ± 4.47	91.20 ± 4.77	91.75 ± 2.72	46.43 ± 13.14	81.21 ± 2.84	81.19 ± 3.72	89.88 ± 5.10	67.82 ± 11.34	74.60 ± 5.99	**93.66 ± 3.40**

TABLE VII

CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE SA DATASET

No.	SSRN (2018)	CEGCN (2021)	WFCG (2022)	DGFNet (2023)	GTFN (2023)	PSFormer (2024)	HGCN-MHF (2024)	GTCFN (2025)	NESSGGCN (2025)	GSDG-Net (Ours)
1	99.29 ± 0.87	99.40 ± 1.61	99.56 ± 0.87	99.31 ± 1.65	98.90 ± 1.21	99.32 ± 0.73	**99.93 ± 0.19**	97.06 ± 2.90	99.68 ± 0.60	99.07 ± 1.58
2	99.42 ± 0.95	99.92 ± 0.25	**100.00 ± 0.01**	96.91 ± 8.83	99.70 ± 0.58	99.86 ± 0.25	**100.00 ± 0.00**	98.04 ± 1.91	99.75 ± 0.61	99.91 ± 0.25
3	99.58 ± 0.43	**100.00 ± 0.00**	99.96 ± 0.12	96.84 ± 5.87	99.88 ± 0.16	99.13 ± 1.78	**100.00 ± 0.00**	96.79 ± 5.45	86.04 ± 11.35	99.78 ± 0.61
4	99.12 ± 1.36	99.87 ± 0.12	99.83 ± 0.09	96.96 ± 3.65	98.92 ± 0.89	**100.00 ± 0.00**	99.20 ± 1.62	98.40 ± 1.14	98.73 ± 2.77	99.70 ± 0.54
5	97.28 ± 1.64	97.78 ± 1.11	97.77 ± 1.46	94.18 ± 5.53	96.41 ± 1.45	97.03 ± 3.75	97.60 ± 2.95	80.53 ± 5.71	**98.46 ± 0.81**	98.19 ± 1.15
6	99.42 ± 0.75	99.93 ± 0.08	99.79 ± 0.52	97.24 ± 1.69	99.39 ± 0.63	99.85 ± 0.18	99.52 ± 0.70	99.69 ± 0.35	**99.96 ± 0.08**	99.36 ± 1.00
7	98.87 ± 1.53	99.94 ± 0.13	99.96 ± 0.05	96.04 ± 7.10	99.48 ± 0.82	99.90 ± 0.10	**100.00 ± 0.00**	99.09 ± 1.32	99.97 ± 0.04	99.88 ± 0.18
8	76.29 ± 12.09	81.11 ± 7.87	82.06 ± 9.07	80.08 ± 8.72	76.43 ± 7.88	72.60 ± 19.87	88.62 ± 7.72	89.37 ± 3.26	85.95 ± 13.56	**93.92 ± 3.53**
9	97.69 ± 2.56	99.98 ± 0.06	99.96 ± 0.09	95.85 ± 5.93	99.73 ± 0.41	99.99 ± 0.03	**100.00 ± 0.00**	86.57 ± 12.14	99.78 ± 0.21	99.84 ± 0.36
10	92.06 ± 4.38	93.96 ± 5.58	94.39 ± 3.77	94.54 ± 3.63	93.53 ± 2.49	91.20 ± 7.17	97.14 ± 3.01	92.32 ± 4.54	94.81 ± 3.91	**97.25 ± 3.10**
11	99.69 ± 0.42	99.82 ± 0.24	99.96 ± 0.06	99.41 ± 1.08	99.71 ± 0.35	99.03 ± 0.78	**99.99 ± 0.03**	96.84 ± 2.42	99.11 ± 1.00	99.96 ± 0.11
12	96.48 ± 3.21	99.96 ± 0.09	**100.00 ± 0.00**	97.43 ± 3.53	98.89 ± 0.76	99.42 ± 1.72	**100.00 ± 0.00**	98.86 ± 1.44	99.29 ± 1.97	99.97 ± 0.06
13	99.58 ± 0.83	**99.82 ± 0.29**	99.65 ± 0.54	94.39 ± 4.28	97.87 ± 3.64	99.48 ± 0.46	99.69 ± 0.49	99.61 ± 0.42	99.38 ± 0.82	99.39 ± 0.93
14	97.09 ± 3.89	99.42 ± 0.55	99.08 ± 1.41	96.40 ± 3.06	98.10 ± 1.22	96.57 ± 2.42	99.10 ± 0.75	98.78 ± 1.46	99.05 ± 1.07	**99.66 ± 0.30**
15	79.14 ± 6.99	91.21 ± 7.37	90.96 ± 5.68	82.73 ± 10.56	90.13 ± 4.06	86.56 ± 12.45	85.35 ± 28.66	89.98 ± 8.02	90.99 ± 5.81	**97.01 ± 2.71**
16	98.39 ± 1.55	98.29 ± 2.77	98.57 ± 2.85	95.14 ± 2.90	97.96 ± 2.53	96.36 ± 3.19	99.20 ± 1.69	89.78 ± 3.14	95.54 ± 3.50	**99.69 ± 0.41**
OA	90.90 ± 2.11	94.28 ± 1.29	94.48 ± 2.20	91.10 ± 1.88	92.80 ± 1.56	91.47 ± 3.11	95.26 ± 3.30	92.51 ± 2.54	94.65 ± 2.61	**97.91 ± 0.85**
AA	95.59 ± 0.44	97.40 ± 0.41	97.46 ± 0.85	94.59 ± 1.31	96.57 ± 0.71	96.02 ± 0.98	97.83 ± 1.49	94.48 ± 1.89	96.66 ± 1.35	**98.90 ± 0.36**
Kappa	89.89 ± 2.32	93.65 ± 1.43	93.86 ± 2.44	90.12 ± 2.08	92.01 ± 1.72	90.55 ± 3.40	94.71 ± 3.72	91.67 ± 2.80	94.05 ± 2.88	**97.68 ± 0.95**

TABLE VIII

CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE UP DATASET

No.	SSRN (2018)	CEGCN (2021)	WFCG (2022)	DGFNet (2023)	GTFN (2023)	PSFormer (2024)	HGCN-MHF (2024)	GTCFN (2025)	NESSGGCN (2025)	GSDG-Net (Ours)
1	75.22 ± 7.17	86.61 ± 8.39	92.10 ± 10.31	69.01 ± 18.29	64.56 ± 16.19	76.07 ± 7.03	84.84 ± 9.92	71.45 ± 23.99	86.84 ± 4.18	**93.08 ± 8.57**
2	89.60 ± 6.26	88.47 ± 4.49	89.72 ± 3.68	61.90 ± 10.43	76.42 ± 10.01	81.73 ± 3.42	85.19 ± 2.52	76.85 ± 27.98	92.05 ± 2.49	**92.53 ± 2.24**
3	92.94 ± 10.62	94.22 ± 5.00	92.46 ± 6.98	73.22 ± 11.99	82.13 ± 6.81	84.98 ± 6.39	**96.02 ± 3.51**	73.18 ± 1.90	89.05 ± 8.35	93.84 ± 4.48
4	97.33 ± 2.44	**99.77 ± 0.19**	99.47 ± 0.57	87.41 ± 11.20	93.74 ± 3.44	94.35 ± 5.81	94.96 ± 9.32	95.19 ± 3.09	99.16 ± 1.09	99.69 ± 0.61
5	92.69 ± 4.99	86.85 ± 4.34	86.03 ± 6.20	51.40 ± 15.08	75.35 ± 7.79	91.78 ± 2.57	**94.44 ± 3.76**	69.61 ± 20.72	86.92 ± 5.29	88.39 ± 5.14
6	**95.59 ± 5.54**	90.43 ± 9.65	93.27 ± 8.91	88.70 ± 6.43	86.42 ± 8.95	83.27 ± 7.07	92.04 ± 8.31	79.31 ± 5.49	89.03 ± 8.02	94.05 ± 8.91
7	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	92.00 ± 3.38	96.06 ± 4.07	**100.00 ± 0.00**	**100.00 ± 0.00**	94.79 ± 6.56	**100.00 ± 0.00**	**100.00 ± 0.00**
8	81.25 ± 8.62	**92.60 ± 11.17**	92.50 ± 8.23	69.38 ± 9.92	82.50 ± 7.40	84.06 ± 6.22	91.34 ± 8.55	75.62 ± 3.97	86.25 ± 12.03	92.08 ± 5.99
9	85.82 ± 5.14	90.14 ± 7.64	**93.60 ± 4.25**	75.10 ± 8.48	78.53 ± 8.02	84.01 ± 7.17	82.32 ± 12.96	89.32 ± 4.43	91.64 ± 4.27	91.78 ± 6.84
OA	89.94 ± 1.46	88.41 ± 2.35	89.33 ± 2.27	62.07 ± 6.73	76.39 ± 2.29	85.93 ± 1.98	89.91 ± 1.24	74.57 ± 7.09	89.42 ± 1.83	**91.32 ± 1.99**
AA	90.05 ± 1.96	92.81 ± 1.95	93.38 ± 1.72	74.24 ± 2.94	81.75 ± 2.43	86.69 ± 2.40	91.24 ± 2.69	80.59 ± 4.50	91.22 ± 1.98	**94.05 ± 1.97**
Kappa	86.04 ± 1.99	84.15 ± 3.11	85.35 ± 3.05	52.19 ± 6.65	68.69 ± 2.76	80.71 ± 2.68	86.05 ± 1.68	64.64 ± 10.09	85.53 ± 2.35	**88.09 ± 2.68**

to a significant performance degradation across all datasets. For instance, the OA on the IP and PU datasets drops by 3.50% and 1.74%, respectively, compared to the full model (Case 4). This decline underscores the indispensable role of A_{gsp} in regularizing the dynamic graph learning. In the original spectral domain, superpixels with similar spectral signatures might be distributed in completely different spatial regions. Without the geometric constraint of A_{gsp}, the self-attention

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TABLE IX

ABLATION STUDY ON THE FDSM AND DSSG MODULES, INCLUDING PARAMETER ANALYSIS

Data set	FDSM	DSSG	Params (K)	OA (%)	AA (%)	Kappa (%)
IP	×	×	119.472	88.21 ± 2.24	92.77 ± 1.51	86.62 ± 2.54
	✓	×	119.602	88.43 ± 2.72	92.96 ± 1.52	86.89 ± 3.05
	×	✓	123.600	88.65 ± 2.58	93.27 ± 1.42	87.14 ± 2.90
	✓	✓	123.730	**90.10 ± 2.38**	**93.88 ± 1.36**	**88.76 ± 2.68**
PU	×	×	106.407	93.10 ± 3.40	96.82 ± 1.18	91.12 ± 4.27
	✓	×	106.537	93.44 ± 3.09	96.95 ± 0.96	91.52 ± 3.90
	×	✓	110.535	94.70 ± 3.11	97.01 ± 1.63	93.12 ± 3.95
	✓	✓	110.665	**95.12 ± 2.67**	**97.53 ± 0.84**	**93.66 ± 3.40**
SA	×	×	119.992	96.85 ± 1.40	98.33 ± 0.53	96.50 ± 1.55
	✓	×	120.122	96.95 ± 1.06	98.47 ± 0.44	96.61 ± 1.18
	×	✓	124.120	97.61 ± 0.80	98.57 ± 0.50	97.34 ± 0.89
	✓	✓	124.250	**97.91 ± 0.85**	**98.90 ± 0.36**	**97.68 ± 0.95**
UP	×	×	149.177	89.82 ± 2.54	93.57 ± 1.94	86.08 ± 3.38
	✓	×	149.307	89.93 ± 2.06	93.36 ± 2.08	86.21 ± 2.75
	×	✓	153.305	90.06 ± 2.10	92.61 ± 1.74	86.36 ± 2.83
	✓	✓	153.435	**91.32 ± 1.99**	**94.05 ± 1.97**	**88.09 ± 2.68**

TABLE X

COMPARISON OF FDSM WITH ALTERNATIVE SPECTRAL MODELING STRATEGIES ON FOUR DATASETS

Methods	Metrics	Datasets			
		IP	PU	SA	UP
Conv1D	OA(%)	88.90±2.28	94.17±3.78	97.62±1.22	89.07±2.07
	AA(%)	93.18±1.17	97.20±1.20	98.44±0.69	92.66±1.50
	Param(K)	125.648	112.583	126.168	155.353
BandAttention	OA(%)	88.58±2.58	94.66±2.52	97.10±1.63	88.42±1.62
	AA(%)	93.12±1.47	97.62±0.55	98.48±0.65	92.55±1.35
	Param(K)	127.696	114.631	128.216	157.401
RealFrequency	OA(%)	89.14±2.75	94.81±2.87	97.39±0.95	88.64±1.75
	AA(%)	93.43±1.44	97.08±1.52	98.50±0.58	92.67±1.50
	Param(K)	123.665	110.600	124.185	153.370
FDSM	OA(%)	90.10±2.38	95.12±2.67	97.91±0.85	91.32±1.99
	AA(%)	93.88±1.36	97.53±0.84	98.90±0.36	94.05±1.97
	Param(K)	123.730	110.665	124.250	153.435

TABLE XI

EFFECTIVENESS OF DIFFERENT MECHANISMS IN DYNAMIC GRAPH CONSTRUCTION

Data set	Case	*topk*	*pos*	A_{gsp}	OA	AA	Kappa
IP	1	×	✓	✓	87.69 ± 2.13	91.91 ± 1.31	86.00 ± 2.35
	2	✓	×	✓	88.96 ± 2.70	93.38 ± 1.39	87.49 ± 3.03
	3	✓	✓	×	86.60 ± 2.71	92.31 ± 1.42	84.83 ± 3.03
	4	✓	✓	✓	**90.10 ± 2.38**	**93.88 ± 1.36**	**88.76 ± 2.68**
PU	1	×	✓	✓	90.99 ± 2.48	94.06 ± 1.34	88.34 ± 3.06
	2	✓	×	✓	94.82 ± 2.31	97.18 ± 0.87	93.26 ± 2.95
	3	✓	✓	×	93.38 ± 2.64	95.90 ± 1.24	91.42 ± 3.34
	4	✓	✓	✓	**95.12 ± 2.67**	**97.53 ± 0.84**	**93.66 ± 3.40**
SA	1	×	✓	✓	93.32 ± 2.13	96.73 ± 1.03	92.58 ± 2.36
	2	✓	×	✓	97.46 ± 0.92	98.79 ± 0.39	97.17 ± 1.02
	3	✓	✓	×	97.33 ± 0.84	98.57 ± 0.38	97.03 ± 0.93
	4	✓	✓	✓	**97.91 ± 0.85**	**98.90 ± 0.36**	**97.68 ± 0.95**
UP	1	×	✓	✓	90.82 ± 3.07	93.56 ± 2.26	87.43 ± 4.09
	2	✓	×	✓	90.83 ± 1.21	93.25 ± 2.57	87.39 ± 1.62
	3	✓	✓	×	90.46 ± 2.06	92.47 ± 1.68	86.93 ± 2.80
	4	✓	✓	✓	**91.32 ± 1.99**	**94.05 ± 1.97**	**88.09 ± 2.68**

mechanism is prone to establishing spurious long-range connections, which introduce irrelevant noise and lead to the "over-smoothing" of node features. By injecting A_{gsp} as a structural bias, GSDG-Net effectively prunes the search space, ensuring that dynamic edge evolution is performed within a geologically and spatially plausible candidate domain. Furthermore, Case 1 and Case 2 demonstrate that neglecting sparsity (*topk*) or position information (*pos*) also hinders the OA. Specifically, the *topk* strategy acts as a topological noise filter, while the *pos* encoding prevents the graph from losing its relative spatial layout in the non-Euclidean space. The fact that Case 4 consistently achieves the peak performance

TABLE XII

EFFECTIVENESS ANALYSIS OF THE DUAL-BRANCH FUSION ARCHITECTURE

Methods	Metrics	Datasets			
		IP	PU	SA	UP
w/o CNN Branch	Param(K)	107.218	94.153	107.738	136.923
	OA(%)	79.86 ± 2.65	72.40 ± 2.88	91.48 ± 0.99	77.52 ± 3.06
w/o GNN Branch	Param(K)	60.578	47.513	61.098	90.283
	OA(%)	83.78 ± 3.33	89.07 ± 3.53	92.88 ± 2.04	90.79 ± 3.34
Dual Branch	Param(K)	**123.730**	**110.665**	**124.250**	**153.435**
	OA(%)	**90.10 ± 2.38**	**95.12 ± 2.67**	**97.91 ± 0.85**	**91.32 ± 1.99**

TABLE XIII

COMPARISON OF OA (%) WITH DIFFERENT METRIC LEARNING METHODS

Method	**IP**	**PU**	**SA**	**UP**
Cosine	89.28 ± 2.50	94.57 ± 2.77	97.65 ± 1.13	91.07 ± 2.89
RBF	88.51 ± 2.43	94.24 ± 3.35	97.74 ± 1.25	90.91 ± 2.57
Manhattan	88.85 ± 2.82	94.42 ± 3.08	97.52 ± 1.32	90.29 ± 1.19
Mahalanobis	88.89 ± 2.41	93.83 ± 3.31	97.74 ± 1.12	90.15 ± 0.45
Superpixel Attention	**90.10 ± 2.38**	**95.12 ± 2.67**	**97.91 ± 0.85**	**91.32 ± 1.99**

across OA, AA, and Kappa confirms that the synergy of sparsity, positional awareness, and geometric priors is essential for constructing the most discriminative graph structure for complex HSI scenes.

4) Ablation Experiments on the Dual-Branch Architecture: To investigate the necessity of the hybrid design, we compare our dual-branch framework against single-branch variants. The "w/o CNN branch" removes the depthwise separable convolution path, relying solely on graph-based features, while "w/o GNN branch" uses only CNN-extracted local features. Results are listed in Table XII.

Performance drops significantly in both ablated models, revealing their complementary roles. On IP, where agricultural fields are small and boundaries intricate, removing the CNN branch causes OA to fall from 90.10% to 79.86%, indicating that the graph alone cannot recover fine spatial details. Conversely, omitting the GNN branch degrades performance on PU and SA (e.g., −6.05% on PU), as large homogeneous regions require long-range semantic consistency. The modest gain on UP (+ 0.53%) reflects its simpler scene structure. These results indicate that the CNN branch brings a larger performance gain because it captures high-frequency local patterns that provide more direct discriminative information, while the GNN branch mainly models complementary low-frequency regional semantics and long-range contextual relationships. Although their contributions differ in magnitude, the fusion of the two branches is still important for complex land-cover classification.

B. Analysis of Content Similarity Measurement

The construction of the graph structure relies heavily on how the similarity between nodes (superpixels) is defined. To explore the optimal strategy for measuring content similarity, we conducted comparative experiments using four traditional metric learning methods—cosine similarity, radial basis function (RBF), Manhattan distance, Mahalanobis distance, and superpixel attention mechanism. The results are reported in Table XIII.

Among the traditional fixed metrics, *cosine similarity* generally yields competitive performance (e.g., 94.57% OA on

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TABLE XIV
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE XUZHOU DATASET UNDER THE SETTING OF TEN LABELED SAMPLES PER CLASS

Metric	SSRN	CEGCN	WFCG	DGFNet	GTFN	PSFormer	HGCN-MHF	GTCFN	NESSGGCN	(GSDG-Net) Ours
OA	85.52±2.80	93.60±1.17	95.38±1.07	84.73±4.53	93.15±1.48	74.14±3.16	95.18±1.32	60.78±19.34	92.64±2.18	**95.92±0.84**
AA	87.88±1.42	97.31±1.31	97.89±0.87	84.87±3.63	96.63±0.76	84.70±1.08	97.32±0.37	65.77±15.85	93.40±1.61	**98.32±0.54**
Kappa	82.06±3.29	92.02±1.44	94.20±1.33	80.95±5.46	92.40±1.64	70.89±3.36	94.35±1.11	52.52±20.45	90.72±2.71	**94.87±1.05**
Params	133.59	196.15	**132.65**	4825.07	169.29	482.24	155.78	330.77	407.62	153.96

PU). This suggests that for hyperspectral data, the directional consistency of spectral vectors is often more discriminative than magnitude-sensitive Euclidean-based metrics like RBF or Manhattan distance. However, these static metrics rely on predefined geometric rules and lack the flexibility to adapt to complex data distributions.

In contrast, *superpixel attention* mechanism achieves the highest accuracy across all four datasets (e.g., 97.91% OA on SA and 95.12% OA on PU). Unlike fixed metrics, the superpixel attention mechanism maps features into a latent subspace via learnable linear projections (W_q and W_k), allowing the model to dynamically optimize the similarity measurement during training. This adaptive capability enables the graph to capture more semantic and task-specific relationships between nodes, thereby significantly enhancing classification performance.

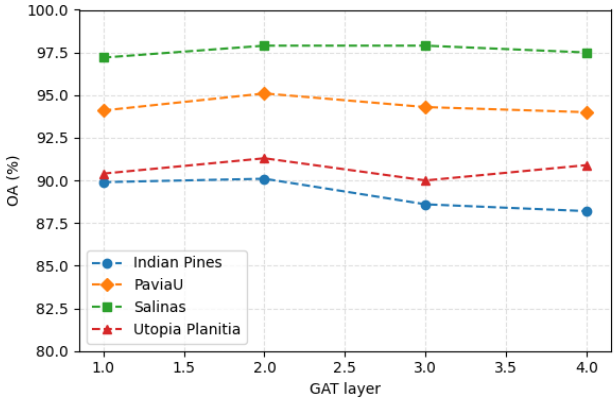


Fig. 12. Sensitivity analysis of the model performance (OA) with respect to the number of GAT layers.

C. Generality and Applicability of GSDG-Net

To further evaluate the generality and applicability of GSDG-Net, we additionally conducted experiments on the Xuzhou dataset, which was collected by an airborne HYSPEX hyperspectral camera over a peri-urban area. This dataset contains both farmland and urban regions, and has a very high spatial resolution of 0.73 m/pixel, making classification more challenging due to its complex spatial structures and mixed land-cover categories. Following the same few-shot setting as in the main experiments, ten labeled samples were randomly selected from each class for training, and the remaining samples were used for testing. In this experiment, the fusion coefficient was set to $\lambda = 0.5$, assigning a relatively larger weight to the graph branch, and the superpixel scale was set to $S = 100$. This setting suggests that the dynamic graph branch plays an important role in modeling complex semantic relationships in such heterogeneous scenes. The remaining settings were kept consistent with those in the main experiments. The corresponding results are reported in Table XIV, further demonstrating the adaptability of the proposed method to high-resolution scenes with diverse spatial characteristics.

D. Impact of GNN Depth

We further investigated the influence of the GNN depth on classification performance by varying the number of graph attention layers from 1 to 4. Fig. 12 illustrates the trends in OA for the four datasets. It can be observed that the performance generally improves as the model depth increases from 1 to 2, reaching a peak at $L = 2$ for the IP, PU, and UP datasets. This improvement indicates that a 2-layer structure effectively aggregates information from 2-hop neighbors, expanding the receptive field to capture broader contextual information. However, stacking more layers ($L > 2$) leads to a decline in performance (e.g., OA drops from 90.10% to 88.21% on IP). This phenomenon is attributed to the *over-smoothing* problem inherent in GNNs, where repeated aggregation operations cause node representations to become indistinguishable, thereby reducing discriminability. Although the SA dataset shows a marginal gain at three layers, the improvement is negligible compared to the computational cost. Consequently, we set the number of GAT layers to 2 as the default configuration to balance representation power and efficiency.

E. Generalization Performance

To further evaluate the generalization ability of GSDG-Net, we vary the number of labeled training samples per class as [6, 8, 10, 12, 14, 16] and compare the OA results with those of the competing methods on three datasets. The corresponding results are shown in Fig. 13(a)–(c). It can be observed that GSDG-Net delivers the best performance in most training settings across all three datasets. A more noticeable advantage can be found on the IP and PU datasets, particularly when only a small number of labeled samples are available. As the number of training samples increases, GSDG-Net continues to achieve competitive results and remains among the top-performing methods on all datasets. Overall, these results indicate that the proposed method exhibits strong robustness and generalization capability, especially in low-sample scenarios.

F. Model Complexity Analysis

As reported in Table XV, our GSDG-Net achieves favorable computational efficiency across the four datasets. To

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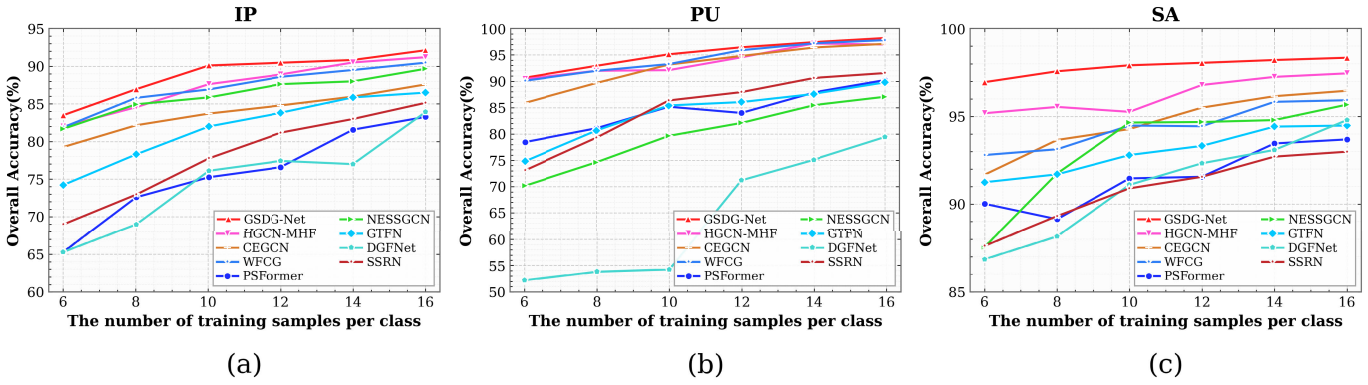


Fig. 13. Evaluation of OA as a function of number of labeled samples per class. (a) IP dataset. (b) PU dataset. (c) SA dataset.

TABLE XV
COMPUTATIONAL COMPLEXITY AND MODEL EFFICIENCY ANALYSIS

Datasets	Metric	SSRN	CEGCN	WFCG	DGFNet	GTFN	PSFormer	HGCN-MHF	GTCFN	NESSGGCN	Ours
IP	Train(S)	60.85	8.47	5.8	25.05	40.94	5.71	6.55	**4.36**	4.49	8.17
	Test(S)	2.18	**0.19**	0.22	2.55	2.52	0.93	0.79	0.22	0.34	0.21
	Flops(G)	0.23	1.61	1.62	**0.03**	0.75	1.66	2.77	3.77	1.76	1.38
	Param(K)	364.168	166.37	**102.42**	4825.974	169.29	482.24	130.5	300.8	377.4	123.73
PU	Train(S)	37.05	45.16	116.72	**27.11**	104.92	64.20	50.83	30.35	27.55	39.51
	Test(S)	5.82	1.43	1.80	10.61	10.40	5.01	5.28	2.18	2.96	**1.33**
	Flops(G)	0.12	13.06	13.23	**0.03**	0.75	12.03	23.58	34.33	14.69	10.83
	Param(K)	216.537	152.85	**89.36**	4825.974	168.84	468.73	112.49	287.29	364.33	110.67
SA	Train(S)	170.84	34.05	40.55	30.31	140.09	36.97	21.55	**15.81**	17.22	68.83
	Test(S)	5.97	**0.99**	1.03	13.43	12.31	5.22	4.79	1.24	1.69	1.01
	Flops(G)	0.24	8.58	8.62	**0.03**	0.75	8.42	14.71	19.97	9.40	7.33
	Param(K)	370.372	166.89	**102.94**	4825.974	169.29	482.76	131.02	301.32	377.92	124.25
UP	Train(S)	183.87	70.43	55.32	**31.51**	50.87	113.15	62.26	51.27	39.46	71.09
	Test(S)	3.85	**2.13**	2.14	17.09	4.70	34.82	29.66	3.41	4.25	2.15
	Flops(G)	0.51	30.03	30.49	**0.03**	0.75	28.09	44.69	59.44	31.74	27.20
	Param(K)	720.345	195.62	**132.13**	4825.974	168.84	511.5	155.26	330.06	407.1	153.44

provide a more practical evaluation of the overall cost in real deployment scenarios, the reported training and testing time is measured over the complete pipeline, including preprocessing steps such as LDA projection, SLIC-based superpixel segmentation, and initial graph construction, in addition to the model execution stage. A closer comparison further reveals different efficiency characteristics among the competing methods. Patch-based models such as SSRN and DGFNet usually have relatively low FLOPs, but their training and testing time are often much longer because the input must be processed in a repeated patch-wise manner, which leads to limited efficiency on large-scale scenes. In contrast, several graph-based or hybrid methods, such as HGCN-MHF, GTCFN, and NESSGGCN, can reduce runtime on some datasets, but this is often accompanied by larger FLOPs or higher parameter counts. As a graph-based method, GSDG-Net also introduces additional structural modeling overhead, yet its parameter count and FLOPs remain at a moderate level compared with these competitors, while the testing time is still competitive across the four datasets. Although the complete pipeline includes preprocessing operations, the overall runtime remains acceptable. Therefore, considering both classification accuracy and deployment cost, GSDG-Net achieves a favorable tradeoff between effectiveness and efficiency in practical HSI classification tasks.

V. CONCLUSION

In this article, we presented GSDG-Net, a learnable superpixel-graph framework for HSI classification. Unlike prior superpixel-based graph methods that rely on preprocessing-defined static adjacency, GSDG-Net jointly optimizes node purification and topology construction in an end-to-end manner. Specifically, we introduced the FDSM module to suppress intrasuperpixel spectral noise by learning complex-valued frequency reweighting, and proposed a centroid-guided self-adaptive graph construction strategy that first forms a geometry-constrained candidate neighborhood and then refines edges via content-aware attention to capture long-range semantic dependencies. Moreover, a dual-branch fusion architecture was adopted to combine global structural reasoning from the dynamic graph with multiscale local spectral–spatial cues from the CNN branch. Extensive experiments on four benchmark datasets demonstrate that GSDG-Net consistently achieves state-of-the-art performance with strong robustness under limited supervision, while maintaining a favorable efficiency–accuracy tradeoff. Nevertheless, the current framework still has some limitations. In particular, its performance may be influenced by the quality of superpixel partitioning, and several hyperparameters still require empirical tuning for different datasets. In future work, we

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plan to explore differentiable superpixel generation and more explicit uncertainty-aware topology learning to further improve robustness in highly mixed or noisy hyperspectral scenes.

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